CHAPTER

ATA 26

FIRE PROTECTION SYSTEM

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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CONTENTS

FIRE PROTECTION, GENERAL

Introduction

The fire protection system includes components throughout the aircraft to provide detection, extinguishing and indication of fire conditions.

Fire Detection System

The fire detection system is designed to sense fire, overheat and smoke conditions.

Refer Figure - Fire Protection System General Locator

The fire detection system consists of:

- Nacelle fire detection system
- Baggage compartment smoke detection system
- Lavatory smoke detection system
- Auxiliary Power Unit (APU) fire detection system.

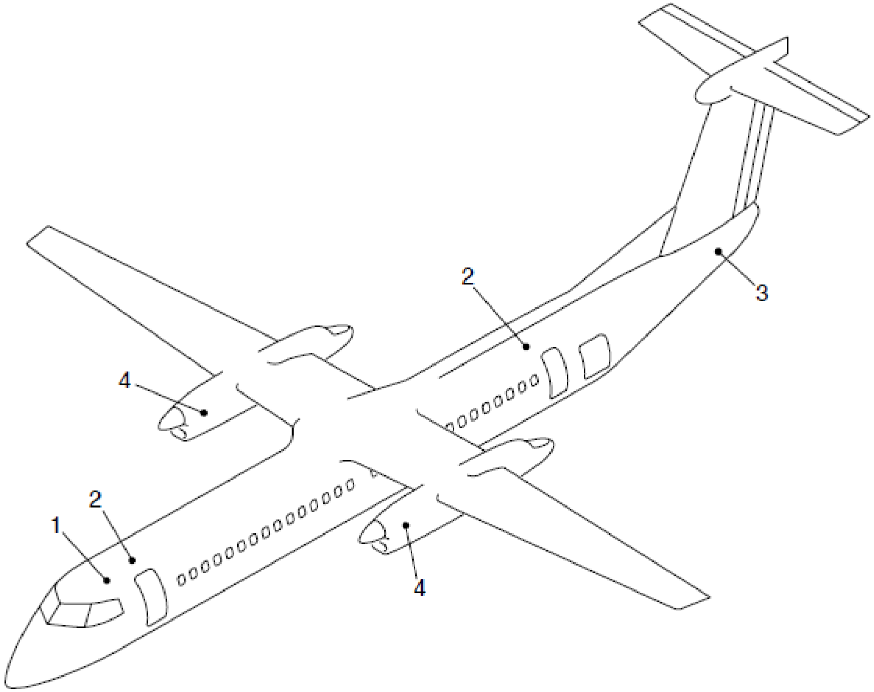
Fire or overheat detection is done by six Advanced Pneumatic Detectors (APD) in the nacelles and one APD in the APU.

6 APD

Smoke detection is done by two smoke detectors located in the aft baggage compartment, one in the forward baggage compartment and one in the lavatory.

- 4 S.D

On aircraft with ModSum 4–458982 and 4–458912 incorporated, the forward baggage compartment has been removed along with the forward HRD fire bottle, fire or smoke detection, indication and the test functions.



LEGEND

1. Lavatory smoke detection system.
2. Baggage compartment smoke detection system.
3. APU fire detection system.
4. Nacelle fire and overheat detection system.

FIGURE - FIRE PROTECTION SYSTEM GENERAL LOCATOR

Fire Extinguishing System

The fire extinguishing system provides components throughout the aircraft for the extinguishing of detected fires.

Refer Figure - Fire Protection Block Diagram

The fire extinguishing system consists of:

- Nacelle fire extinguishing system
- Baggage compartment fire extinguishing system
- APU fire extinguishing system
- Lavatory fire extinguishing system
- Portable hand–operated fire extinguishers.

Refer Figure - Fire Protection System Schematic (Sheet 1 of 2)

Two dual–port fire–extinguishing bottles are installed in the left wing root for engine nacelle fire or overheat suppression. Both bottles can be used to extinguish a fire in either engine.

Fire extinguishing for the baggage compartments is performed by two High Rate Discharge (HRD) fire extinguisher bottles and one Low Rate Discharge (LRD) fire extinguisher bottle. Each baggage compartment has one HRD fire extinguisher bottle; the LRD bottle is shared between the FWD and AFT baggage compartments.

One fire–extinguishing bottle is used to suppress fire and overheat conditions in the APU compartment.

The lavatory waste bin is protected by a single thermally–activated fire–extinguisher.

The flight compartment and passenger compartment are protected by portable hand–operated fire–extinguishers.

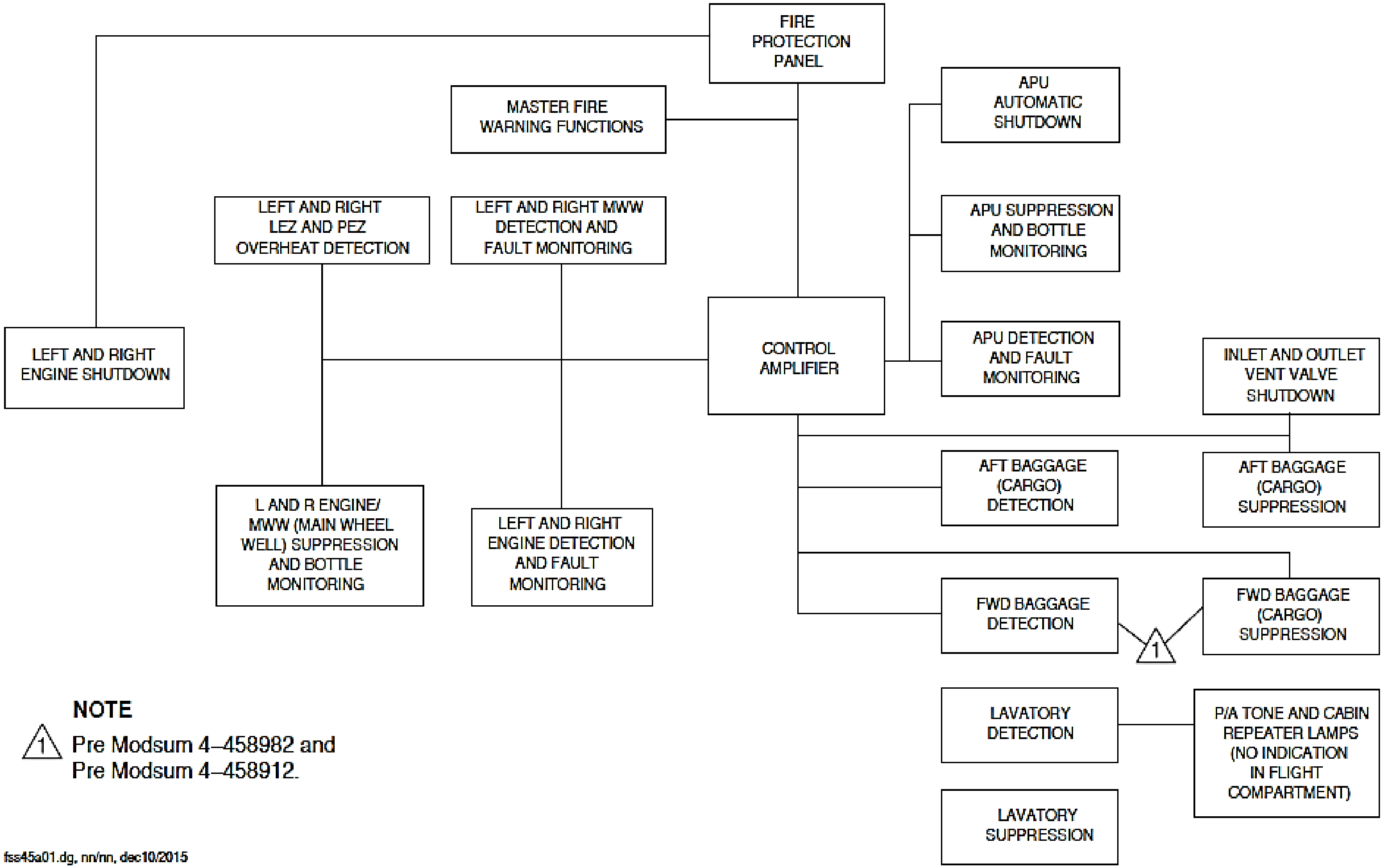


FIGURE - FIRE PROTECTION BLOCK DIAGRAM

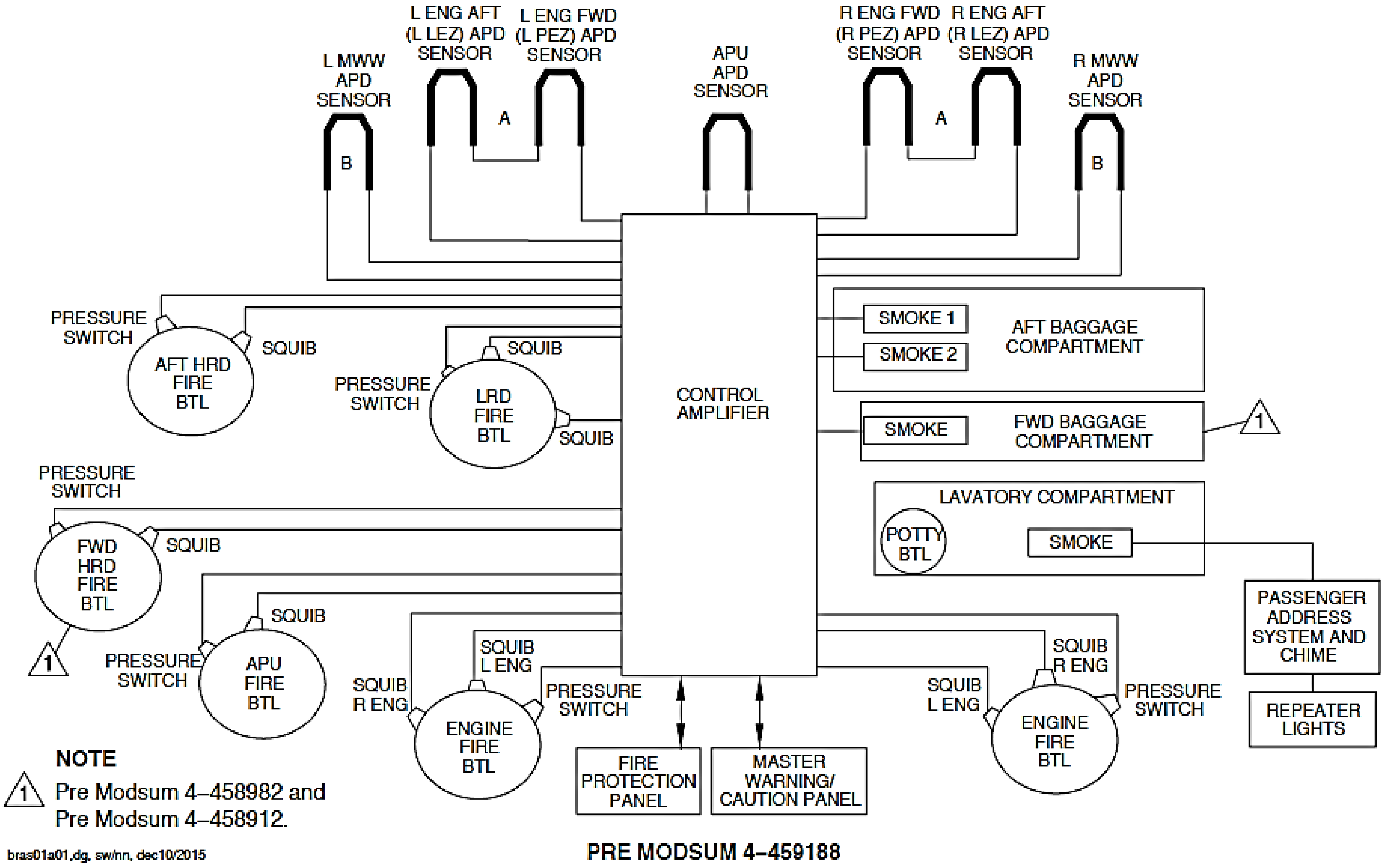


FIGURE - FIRE PROTECTION SYSTEM SCHEMATIC (SHEET 1 OF 2)

Fire Indication and Test functions

Refer Figure - Indications in Flight Compartment for Fire Protection System (Sheet 1 of 2)
Refer Figure - Indications in Flight Compartment for Fire Protection System (Sheet 2 of 2)

Fire or smoke detection is shown on the Fire Protection Panel (FPP), Caution and Warning Panel (CWP) and Glareshield Panel located in the flight compartment.

Indication and test functions are provided on the Fire Protection Panel for:

- Engines
- APU
- Forward and Aft Baggage Compartments.

Indication is provided at the cabin repeater lights for the lavatory smoke and fire detection.

Control Amplifier

Refer Figure - Control Amplifier

The control amplifier of the fire protection system performs fire detection and extinguishing system monitoring and test functions. It is located on the No. 2 equipment panel. – AFT OF AVD. 1 PANEL

The control amplifier does the following:

- Detection and extinguishing monitoring
- Built In Tests (BIT) function
- Controls the advisory lights on the Fire Protection Panel
- Controls the warning lights on the Caution and Warning Panel and Glareshield Panel
- Controls the fire bell chime (optional).

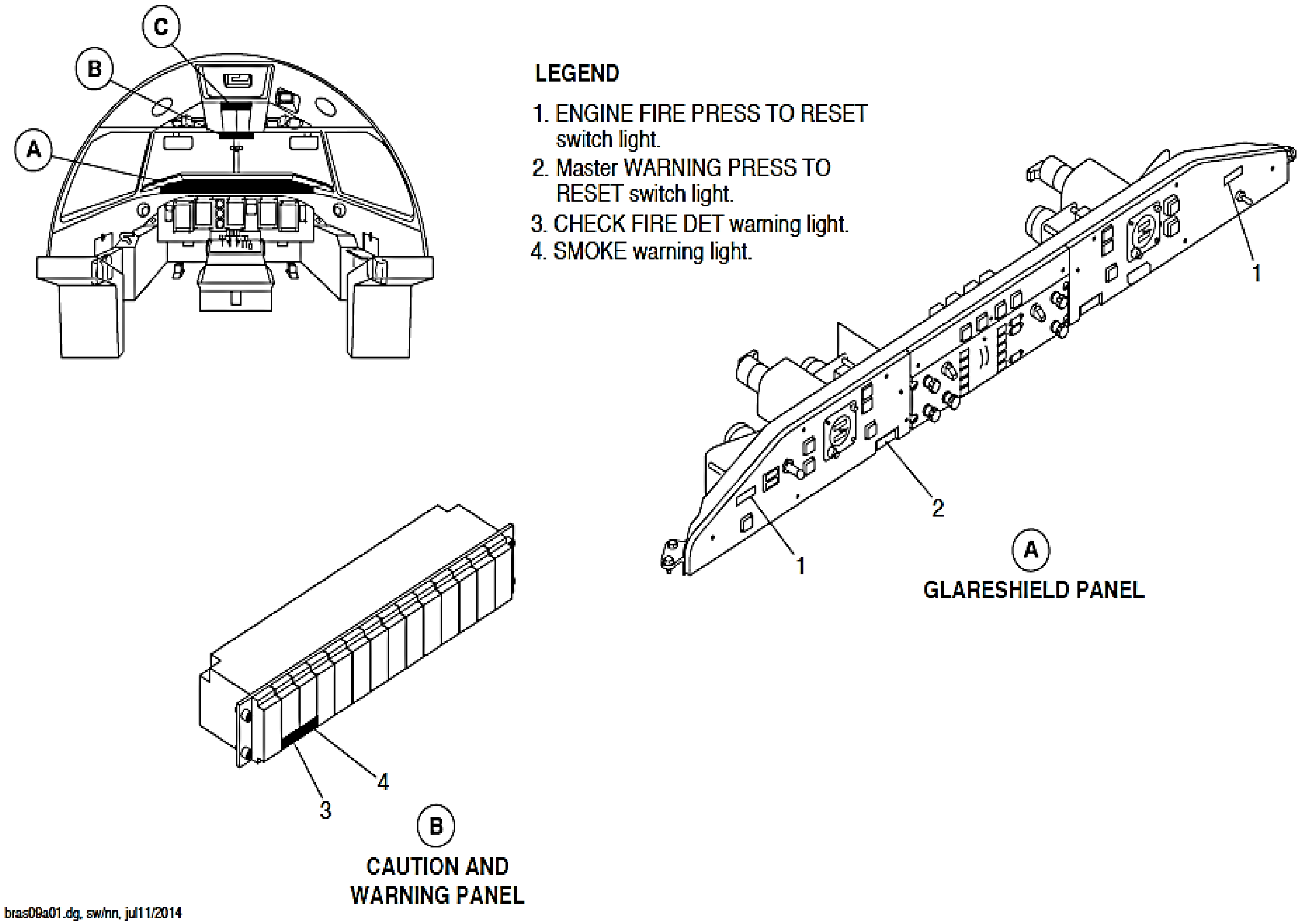


FIGURE - INDICATIONS IN FLIGHT COMPARTMENT FOR FIRE PROTECTION SYSTEM (SHEET 1 OF 2)

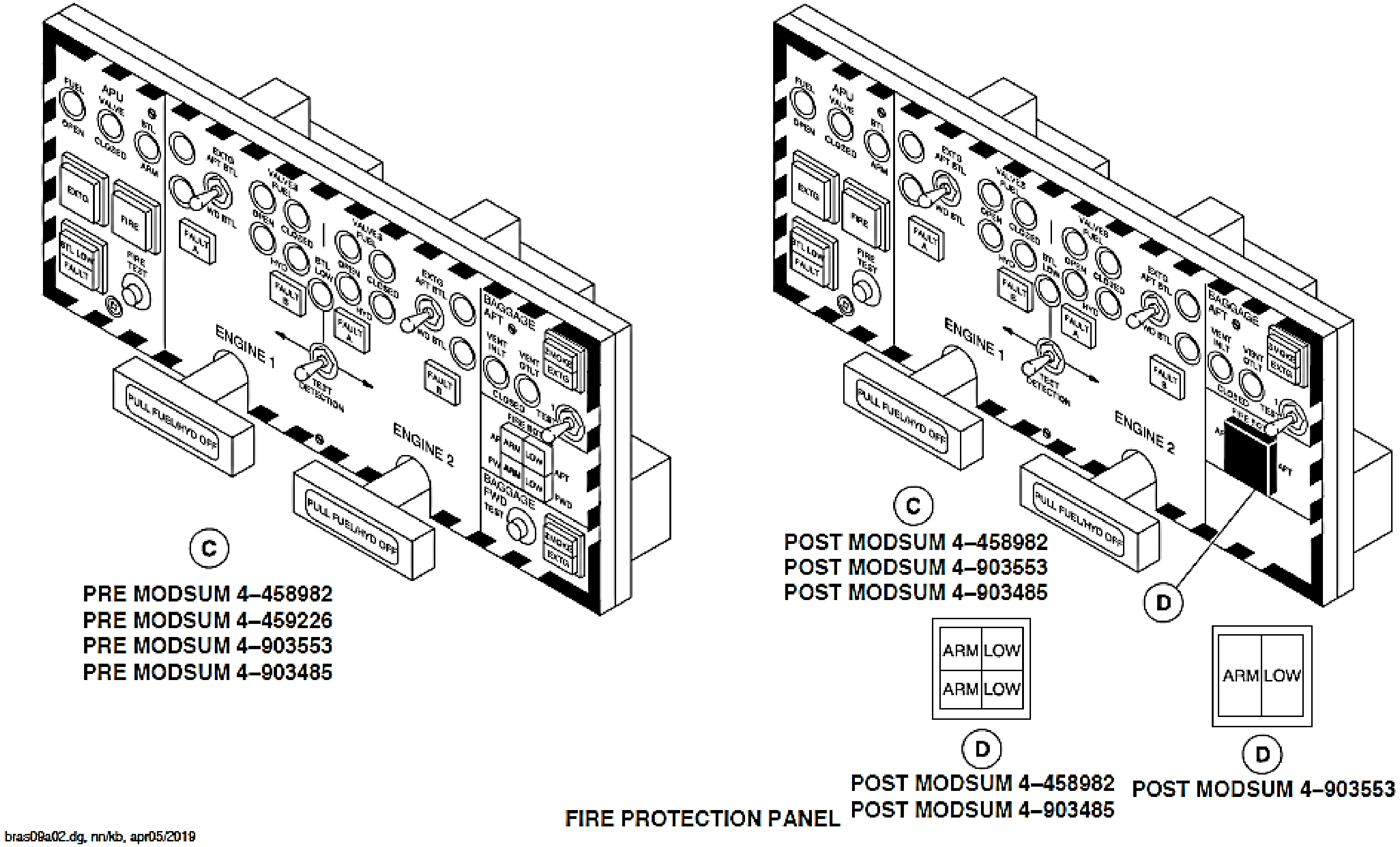


FIGURE - INDICATIONS IN FLIGHT COMPARTMENT FOR FIRE PROTECTION SYSTEM (SHEET 2 OF 2)

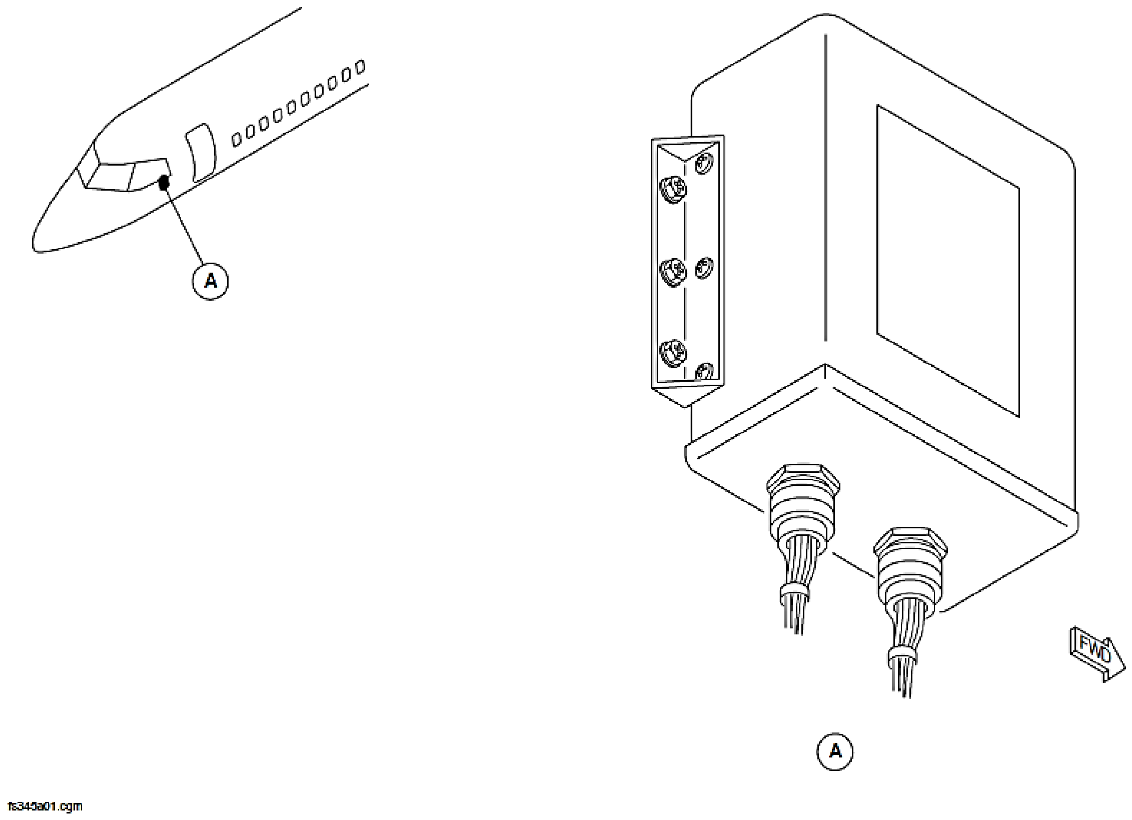


FIGURE - CONTROL AMPLIFIER

NACELLE FIRE DETECTION SYSTEM

Introduction

The nacelle fire detection system supplies indications of fire and overheat conditions in the nacelle fire zones.

General Description

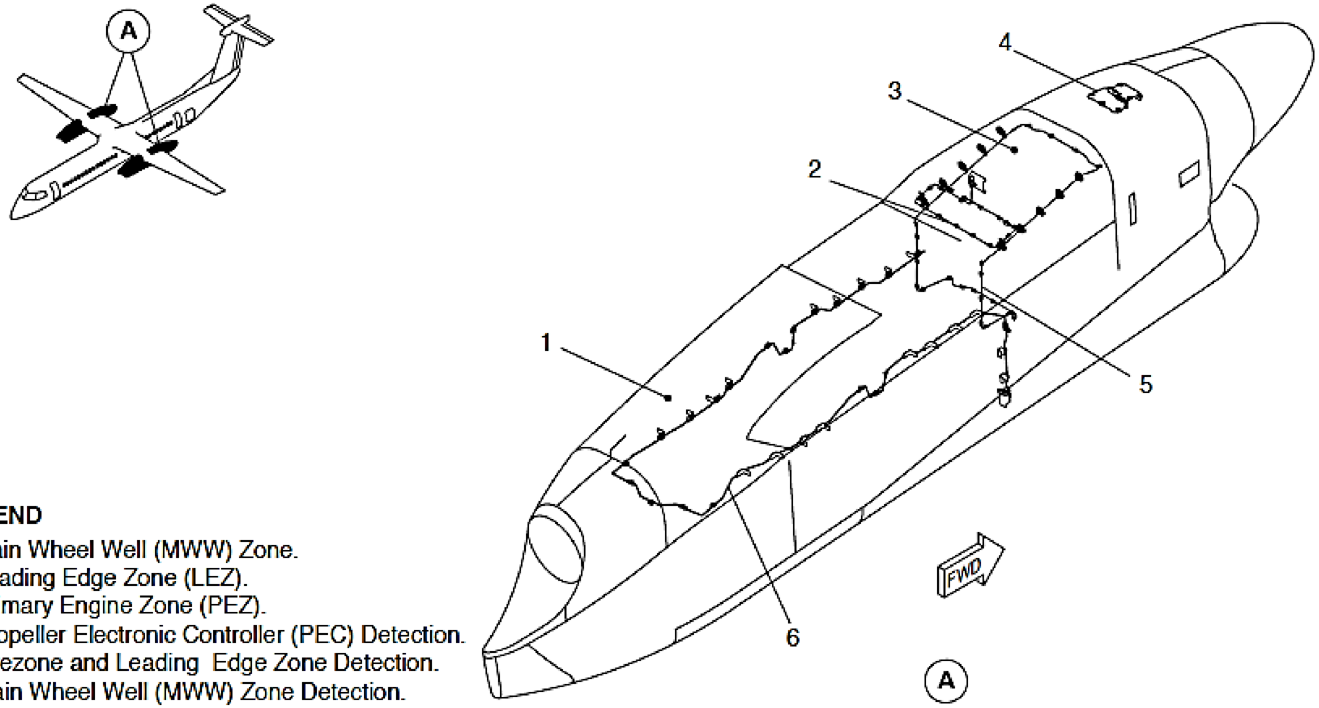
Refer Figure - Advanced Pneumatic Detectors for Nacelle Fire Detection

The fire detection system uses Advanced Pneumatic Detectors (APD) to sense a fire or overheat condition in the nacelles. The fire or overheat condition is shown on the Fire Protection Panel (FPP) in the flight compartment.

Three APD’s are located in each nacelle (left and right). The APD’s or fire detection loops, give fire and overheat detection in the Main Wheel Well (MWW) zone, Leading Edge Zone (LEZ) and Propeller Electronic Controller (PEC) area of the Fire Zone.

The nacelle fire detection system has these components:

- Engine/Leading Edge Zone Advanced Pneumatic Detectors
- Propeller Electronic Control Advanced Pneumatic Detectors
- MLG Zone Advanced Pneumatic Detectors.



LEGEND

1. Main Wheel Well (MWW) Zone.
2. Leading Edge Zone (LEZ).
3. Primary Engine Zone (PEZ).
4. Propeller Electronic Controller (PEC) Detection.
5. Firezone and Leading Edge Zone Detection.
6. Main Wheel Well (MWW) Zone Detection.

NOTE

Left nacelle shown.
Right nacelle similar.

FIGURE - ADVANCED PNEUMATIC DETECTORS FOR NACELLE FIRE DETECTION

Detailed Description

The Advanced Pneumatic Detectors (APD) use gas−filled sensor tubes to monitor heating of the zone.

Refer Figure - Advanced Pneumatic Detector (APD)

Two switches in the APD – the integrity switch and the alarm switch – provide the fault and alarm signals for indication on the Fire Protection Panel (FPP).

Refer Figure - Fire Protection Panel Detail for Nacelle Fire Detection and Test
Refer Figure - Indications for Nacelle Fire

The integrity switch monitors the pressure in the sensor element. If the pressure falls below a minimum rated level, the normally closed integrity switch opens and sends a signal to the control amplifier. The FAULT A or FAULT B light on the FPP comes on.

A fire or overheat condition will cause gas expansion within the APD. When the temperature reaches a set level, the increased gas pressure closes the alarm switch. The APD will generate a fire signal to the appropriate engine PULL FUEL/HYD OFF handle light directly.

When the alarm switch closes, it also sends the signal to the control amplifier and is indicated by:

- PULL FUEL/HYD OFF handle light (red) on the FPP comes on
- CHECK FIRE DET warning light (flashing red) on the Caution and Warning Panel comes on
- Master WARNING PRESS TO RESET light (red) on the Glareshield Panel comes on
- ENGINE FIRE PRESS TO RESET light (flashing red) on the left and right side of the Glareshield Panel comes on
- Audible BELL warning sounds (optional).

Pushing either the left or right ENGINE FIRE PRESS TO RESET indicator turns the audible BELL warning off; Both ENGINE FIRE PRESS TO RESET remain on steady state for the duration of the alarm condition.

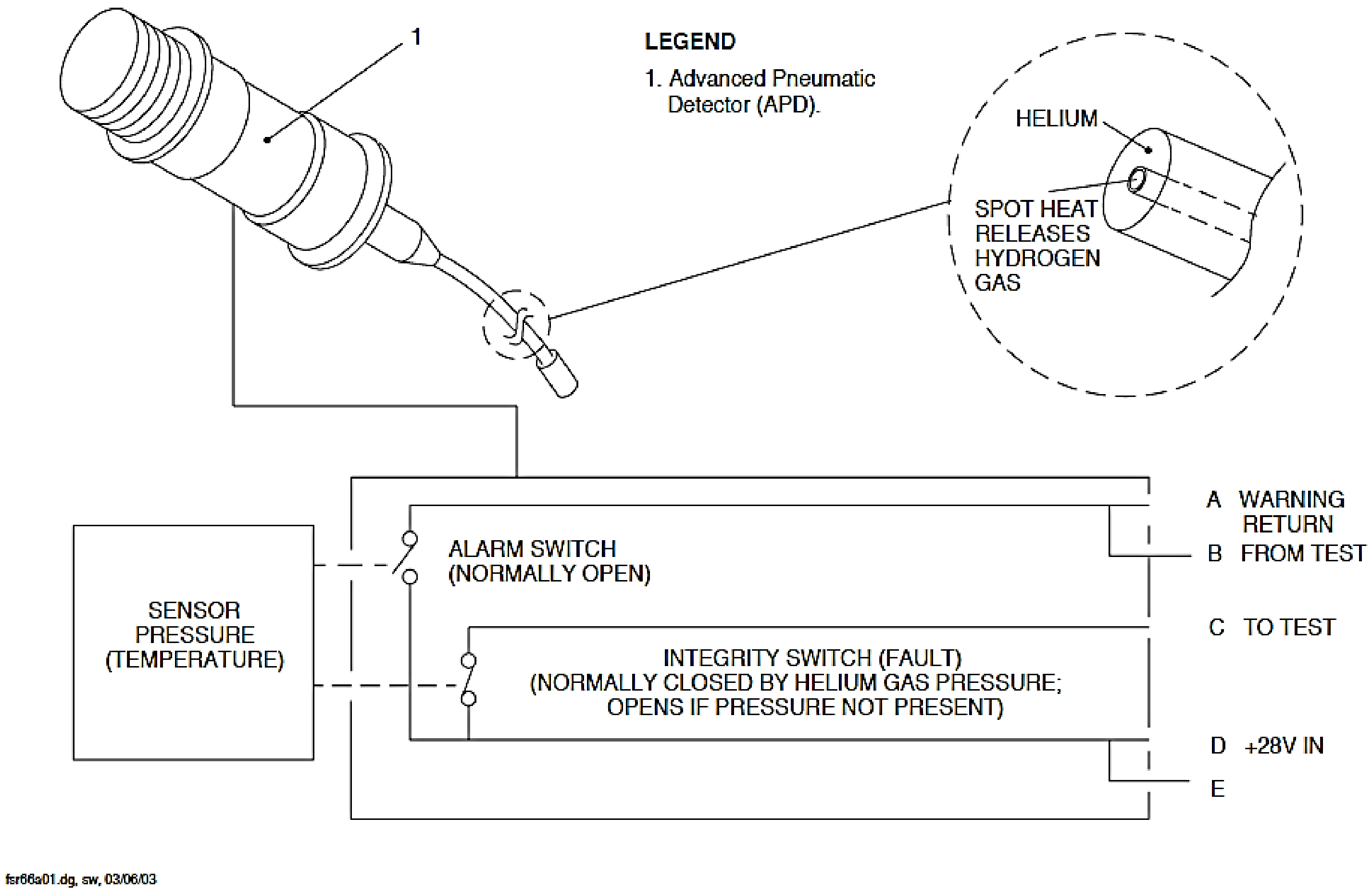
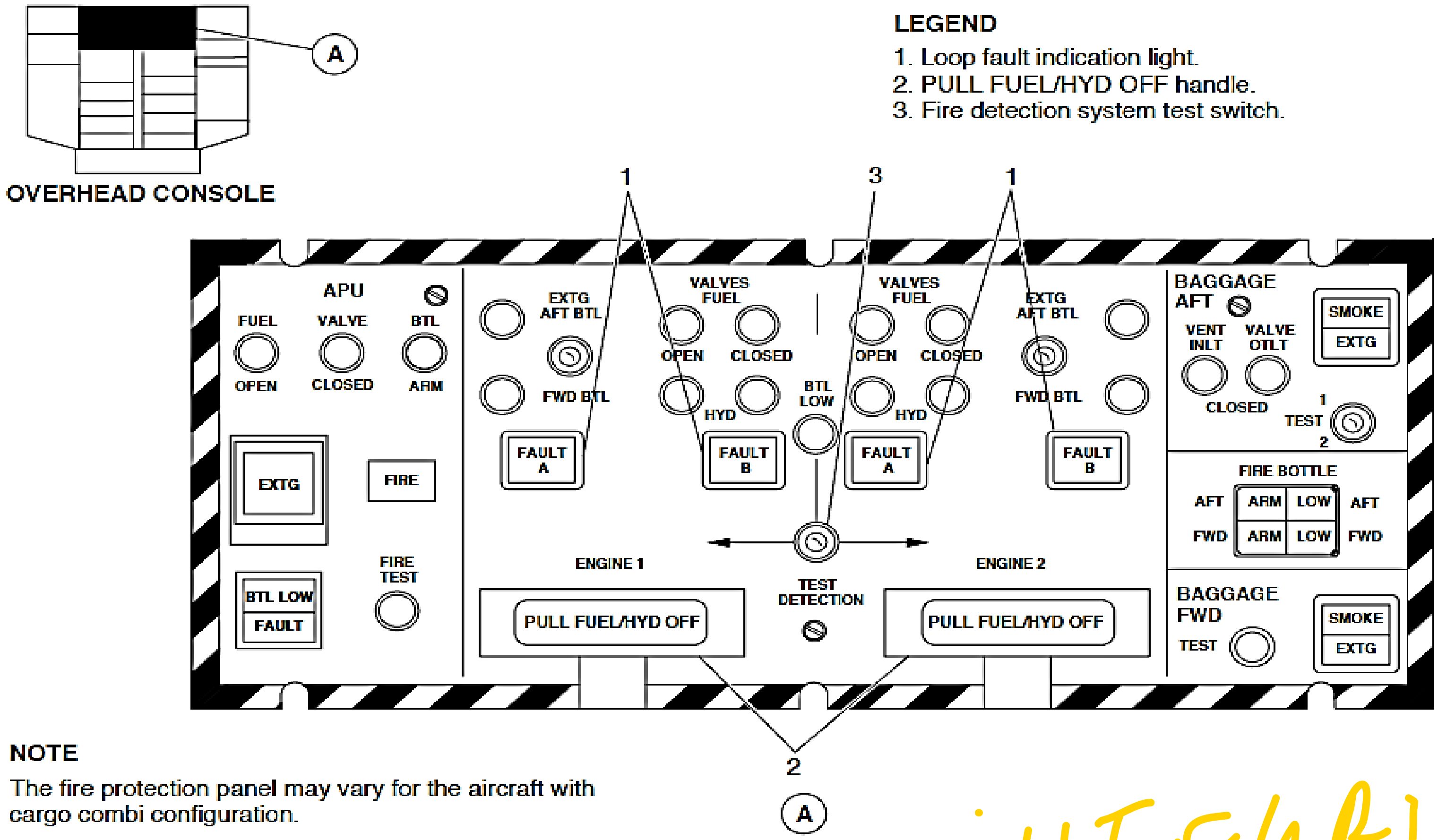


FIGURE - ADVANCED PNEUMATIC DETECTOR (APD)

LEGEND

1. Loop fault indication light.
2. PULL FUEL/HYD OFF handle.
3. Fire detection system test switch.



NOTE

The fire protection panel may vary for the aircraft with cargo combi configuration.

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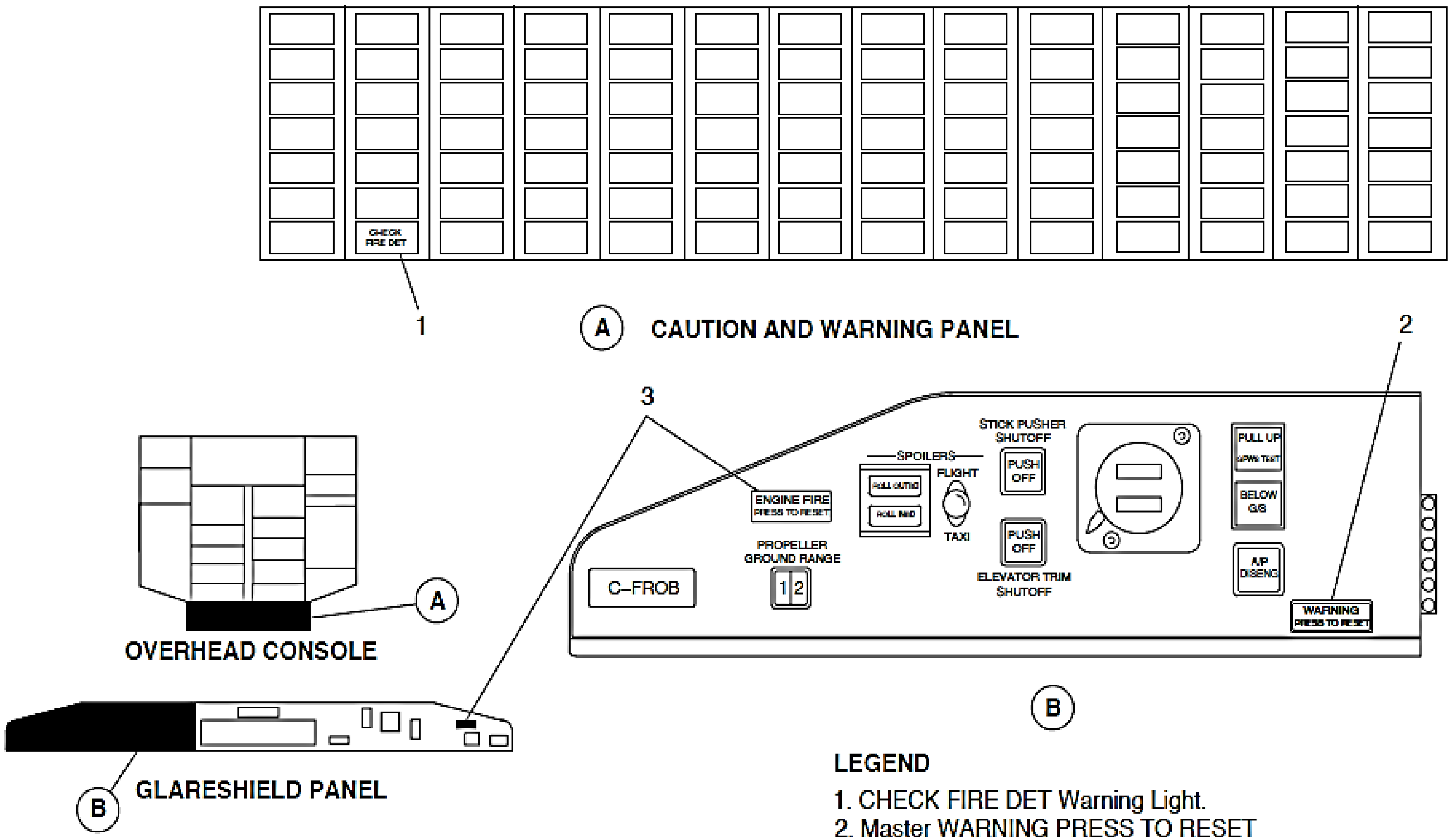
FAULT- INTEGRITY
S/W OPEN

FIGURE - FIRE PROTECTION PANEL DETAIL FOR NACELLE FIRE DETECTION AND TEST



LEGEND

1. CHECK FIRE DET Warning Light.
2. Master WARNING PRESS TO RESET switchlight.
3. ENGINE FIRE PRESS TO RESET Annunciator Switch.

FIGURE - INDICATIONS FOR NACELLE FIRE

Advanced Pneumatic Detectors (APD)

Refer Figure - Integrity and Alarm - APD
Refer Figure - Fault A or B - APD

Advanced Pneumatic Detectors (APD) or fire detection loops are used to sense a fire or overheat condition in the nacelles.

Three APD's are located in each nacelle. They give fire and overheat detection in the Main Wheel Well (MWW) zone, Leading Edge Zone (LEZ) and Propeller Electronic Controller (PEC) area of the Fire Zone.

APD's have a main element body and sensor tube. The main element body is held in place by two P–clamps and the sensor tube is supported by additional clamps and grommets.

APD's are sensor tubes filled with helium gas to monitor for fires. The helium gas is sensitive to changes in temperature; its pressure increases with the increase in temperature. Spot heat releases hydrogen from the central core which increases pressure in the loop.

APD's receive electrical power from the 28 Vdc left essential and right essential bus. Malfunction of either the left or right essential bus will not cause complete loss of fire detection in the left or right nacelles.

Two switches in the APD – the integrity switch and the alarm switch – provide the fault and alarm signals. The fire or overheat condition is shown on the Fire Protection Panel (FPP).

The integrity switch monitors the pressure in the sensor element. If an APD breaks, the loss of pressure in the sensor will open the switch and turn on the FAULT A or FAULT B light on the FPP.

The alarm switch is normally open, and closes when an overheat or fire condition occurs, caused by the gas pressure increase in the APD. The table that follows shows the specified alarm temperature values:

ADVANCED PNEUMATIC DETECTORS (APD) LOCATION	DETECTOR PART NUMBER	ALARM TEMPERATURE °F (°C)	MINIMUM RESET TEMPERATURE °F (°C)
Propeller Electronic Controller (PEC) Zone	8SC1287	282 to 318 (139 to 159)	252 (122)
Main Wheel Well (MWW) Zone	8SC1288	235 to 265 (113 to 129)	210 (99)
Leading Edge Zone (LEZ)	8SC1275	470 to 530 (243 to 277)	420 (216)

The APD alarm switch provides signals to the appropriate engine PULL FUEL/HYD OFF handle light directly. The alarm signals are also processed by the control amplifier and then sent to the FPP.

The sensors in the engine/LEZ and PEC are connected in such a manner that if there is a fault in one of the APD's, a FAULT A indication on the fire protection panel comes on. A fault in the undercarriage zone APD will give a FAULT B indication on the fire protection panel. The CHK FIRE DETECT light and FAULT light indications come on at the same time.

The sensor loops in LEZ and PEC are connected in series with the control amplifier in such a manner that if the loop in PEC fails, the fire zone LEZ loop will still provide fire indications. However, if the LEZ loop fails, the PEC loop will not function as a fire detector. It is therefore possible to receive a FAULT A indication and an engine fire warning concurrently.

System Tests

Refer Figure - Fire Protection Panel Detail for Nacelle Fire Detection and Test
Refer Figure - Indications for Nacelle Fire

To test the nacelle fire detection system, select the (two position, spring−loaded to center) test switch on the Fire Protection Panel to either ENGINE 1 or ENGINE 2.

The nacelle fire detection test indications are as follows:

- Master WARNING PRESS TO RESET light (red) on the Glareshield Panel comes on and flashes
- CHECK FIRE DET warning light (red) on the Caution and Warning Panel comes on and flashes
- Both ENGINE FIRE PRESS TO RESET lights (red) on the Glareshield Panel come on and flash
- PULL FUEL/HYD OFF light (red) on the pull−handle of the Fire Protection Panel come on
- FAULT A and FAULT B lights (amber) on the Fire Protection Panel come on
- FIRE BELL audible warning comes on (optional).

Any nacelle APD will cause the PULL FUEL/HYD OFF handle to illuminate.

The ENGINE FIRE PRESS TO RESET advisory lights will come on only if correct response is obtained from all three nacelle APD’s and the control amplifier is functioning normally.

Pushing either ENGINE FIRE PRESS TO RESET advisory light will silence the fire bell.

Training Information Points

Care must be taken during the installation and removal of the APD. Any damage to the sensor tube can result in a leak of the inert gas which will result in APD sensor failure.

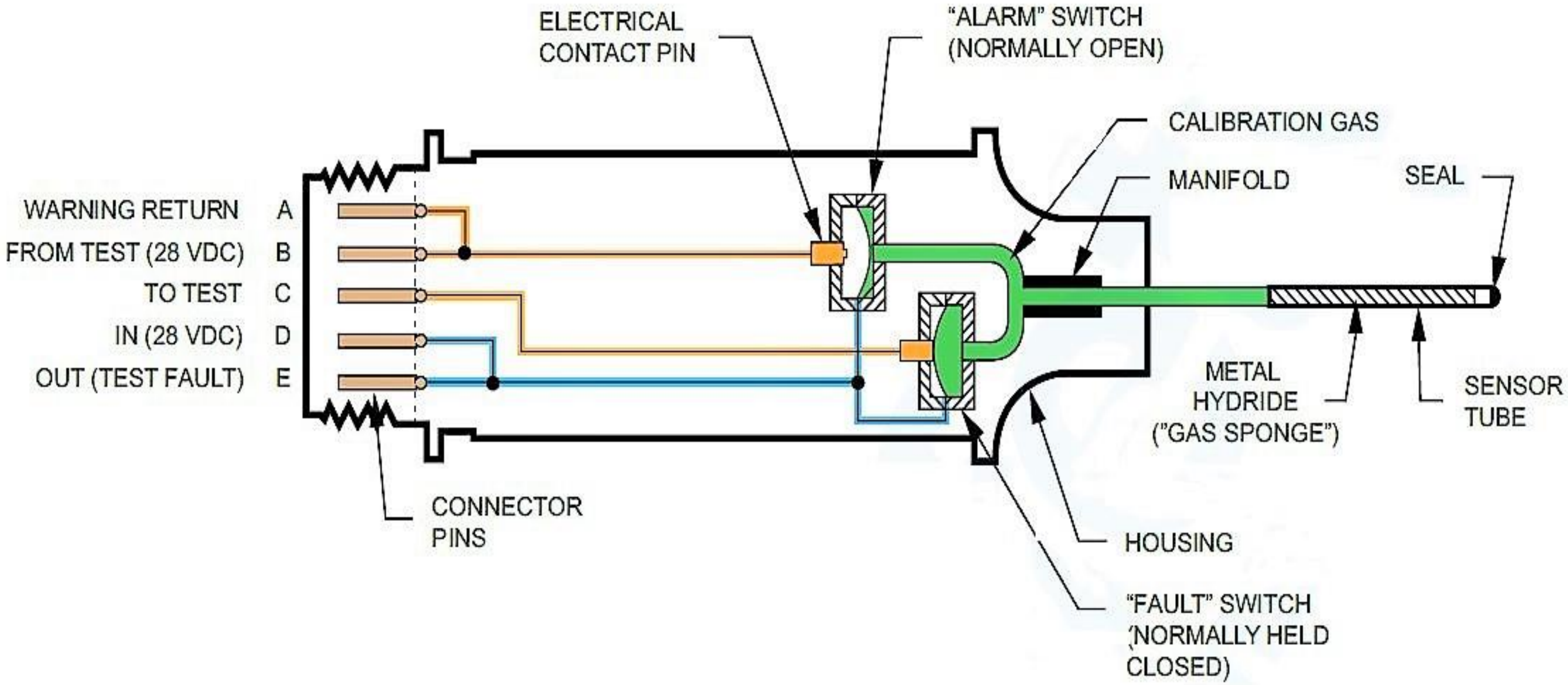


FIGURE - INTEGRITY AND ALARM - APD

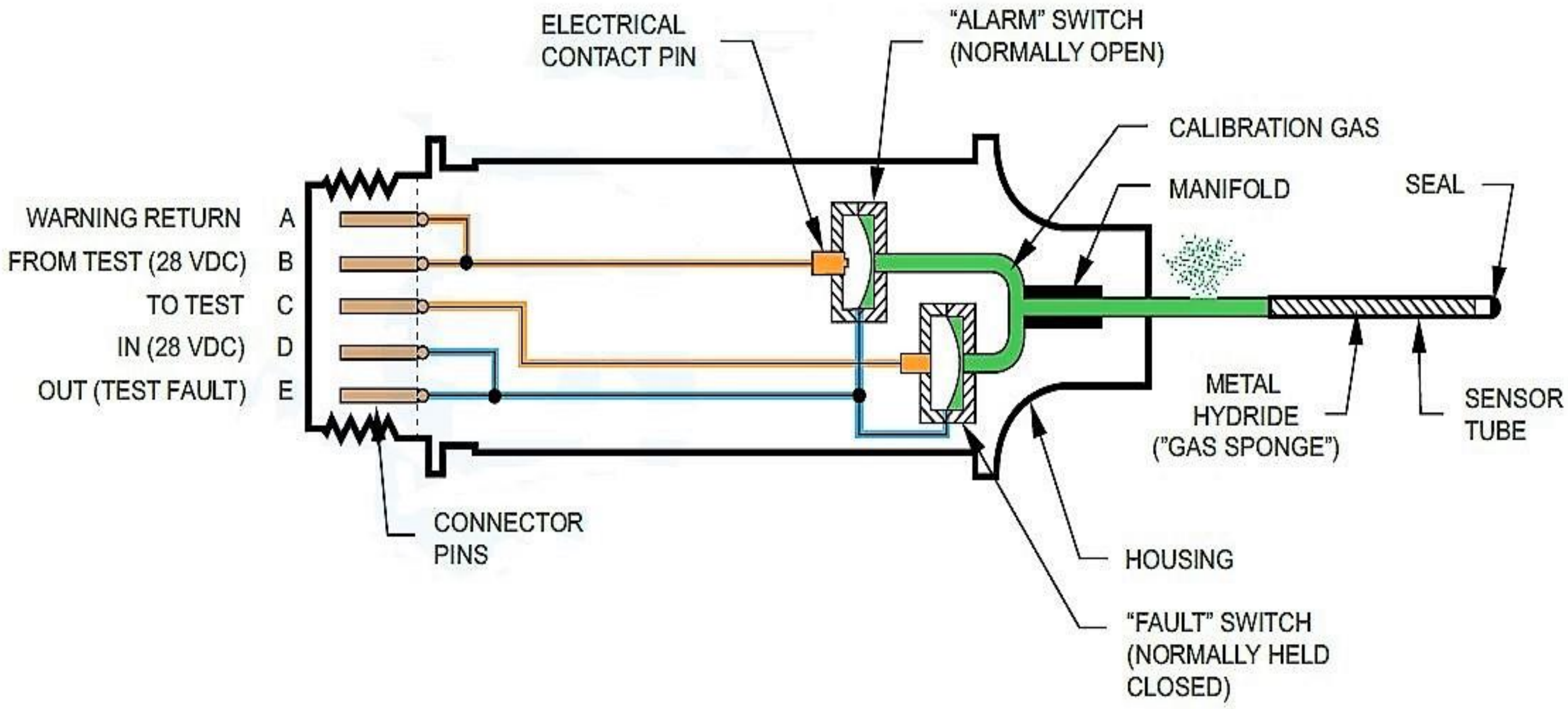


FIGURE - FAULT A OR B - APD

NACELLE FIRE EXTINGUISHING

Introduction

The Nacelle Fire Extinguishing uses two fire–extinguishing bottles for the nacelles fire extinguishing.

General Description

Two dual–port fire–extinguishing bottles are installed FWD and AFT in the left wing root, for engine fire or overheat suppression. Both bottles can be used to extinguish a fire in either engine. Indications in the flight compartment show the flight crew when the bottle pressure is low.

Each bottle is connected through a Check Tee piece with a drain. This gives extinguisher coverage to the right or left nacelle Primary Engine Zone (PEZ), Leading Edge Zone (LEZ) and Main Wheel Well (MWW) zones.

Electrical connections are installed for both explosive squibs and the bottle monitor pressure switch.

Detailed Description

A fire or overheat condition in the left or right nacelle will cause these flight deck indications:

- PULL FUEL/HYD OFF handle light (for related engine, on the Fire Protection Panel)
- CHECK FIRE DET warning light (on the Caution and Warning Panel)
- Master WARNING PRESS TO RESET light on the Glareshield Panel
- ENGINE FIRE PRESS TO RESET lights on the left and right side of the Glareshield Panel
- Audible warning bell.

The PULL FUEL/HYD OFF handle is pulled to close the fuel and hydraulic shut–off valves and arm the FWD and AFT bottles. Two BTL ARM lights on the Fire Protection Panel will come on showing that both bottles have continuity on the squib and pressure switches.

To discharge a bottle, the EXTG switch is set to AFT BTL or FWD BTL position. The selected bottle will discharge its extinguishing into the nacelle area. The bottle pressure switch will open and the BTL LOW light will come on to show the loss of bottle pressure. The BTL ARM lights go off.

Fire Extinguisher Bottles

Refer Figure - Nacelle Fire Extinguisher

The Fire Extinguisher Bottles contain the suppressant necessary to combat nacelle fire or overheat. They are located in the left wing root.

The extinguisher bottles use dual–bridgewire Electro–Explosive Devices (EED) within each cartridge and redundant power lines with separate circuit breakers for reliability.

Bridgewire “A” of all cartridges is taken from one circuit breaker and routed along the left side of the aircraft to the cartridge. Bridgewire “B” of all cartridges is taken from a separate circuit breaker and routed along the right side of the aircraft to the respective cartridge. This arrangement meets the rotor burst requirements and gives redundant firing signals for each cartridge.

On A/C with modsum 4Q126352 incorporated, the electrical connectors are attached with lanyards. These lanyards make sure that the correct connector is connected to the discharge squib during maintenance practices.

The left and right extinguisher bottles are discharged manually from the Fire Protection Panel (FPP) into the Primary Engine Zone (PEZ), Leading Edge Zone (LEZ) and Main Wheel Well (MWW) zones.

The left or right bottle squibs are armed by pulling the PULL FUEL/HYD OFF handle on the FPP. After arming, the pilot can operate one of the extinguisher bottles with the EXTG switch. An electrical signal is sent through the bridgewires, which ignites the EED. When the EED explodes, it ruptures a burst disc and the pressurized bottle then discharges through the connector Check Tee.

The bottle configuration can give two applications of fire suppressant into the zones, in the event that the first attempt was not effective.

The extinguisher bottles are constantly monitored for pressure during the non−alarm state. The bottle pressure switch status is shown by the BTL LOW light on the FPP. This light will come on when either bottle has lost pressure. The PULL FUEL/HYD OFF handle test will show which bottle needs service; the arming light for that bottle will not come on.

A manual exercise key is supplied for ground check of proper pressure switch operation. The key will open the pressure switch contacts, to simulate a depressurized bottle.

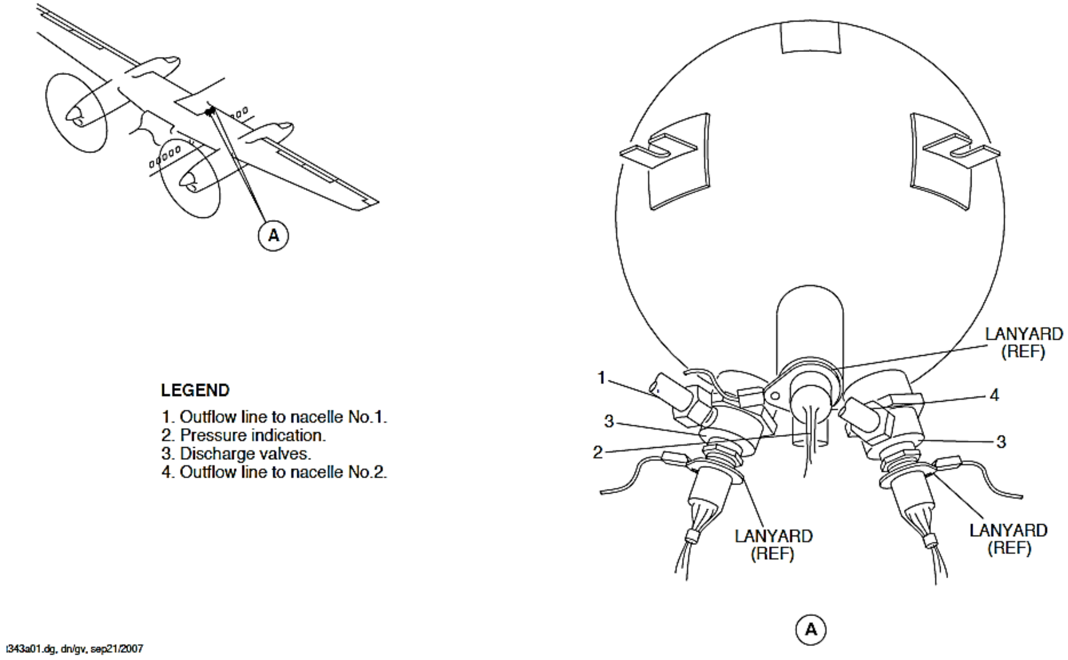


FIGURE - NACELLE FIRE EXTINGUISHER

Figure - Fire Detection - Normal Operation

NOTE

Engine number 1 shown, engine number 2 operations similar.

28 VDC is supplied from the left essential bus via circuit breaker K7 (FIRE DET ENG 1 IND) through disconnects and configuration connectors (removed for clarity) to the APD pin C.

This signal passes through the fault switch that is normally held closed by pressure within the APD and exits on pin D. The voltage from the main wheel well detector is then applied to the fire control amplifier, pin 60. EPZ & PEC integrity switches are connected in series. The voltage that exits pin D on the EPZ is applied to the PEC pin C before being sent to the control amplifier pin 65.

The voltage sensed on pins 60 and 65 confirms the integrity of the nacelle fire detection control loops (APD).

If an APD should lose its pressure or there is a wiring fault, the integrity voltage would not be sensed at the control amplifier. A loss of voltage on pin 60 causes the control amplifier to illuminate the FAULT “B” advisory light on the FPP. A loss of voltage on pin 65 causes the control amplifier to illuminate the FAULT “A” advisory light.

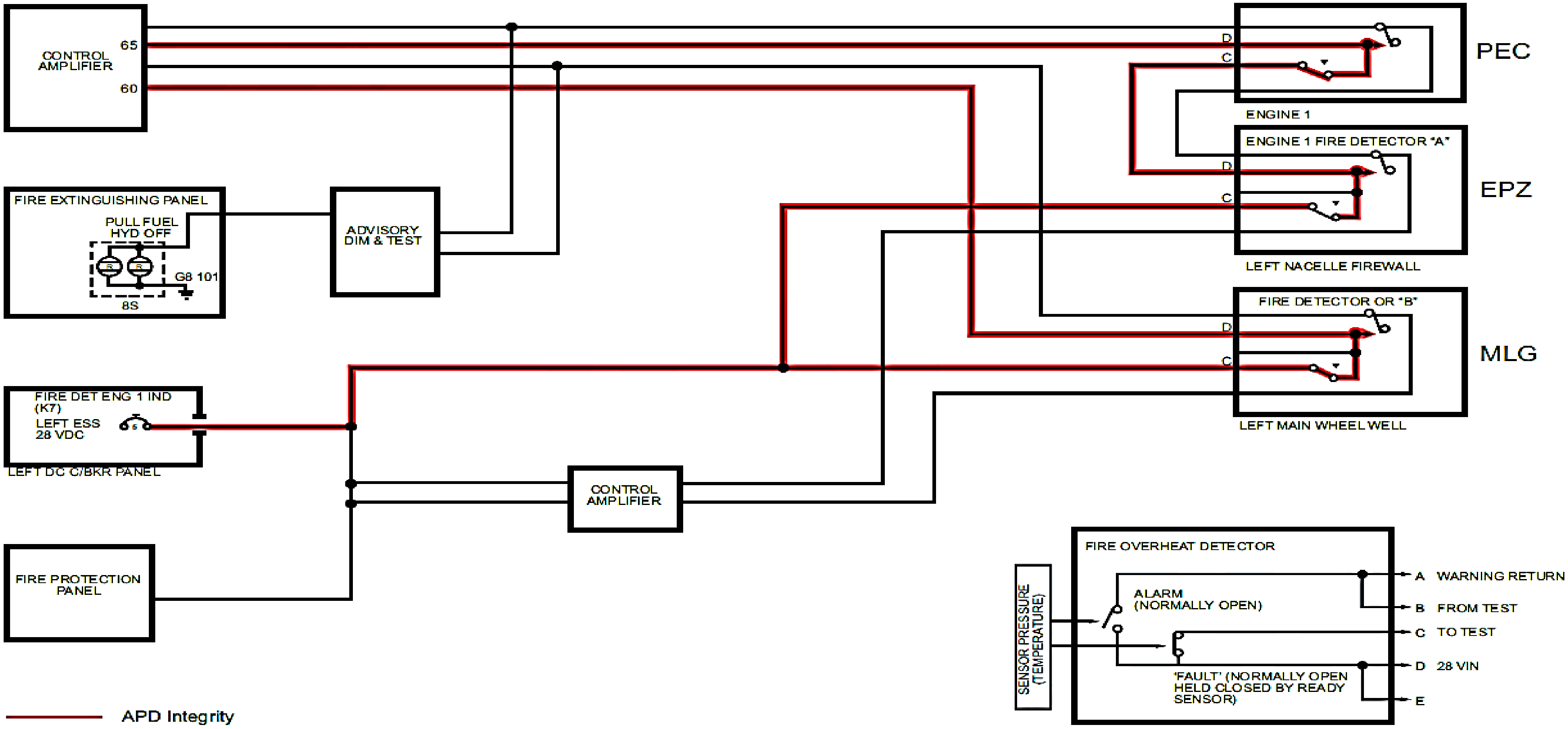


FIGURE - FIRE DETECTION - NORMAL OPERATION

Refer Figure - Fire Overheat Detection - MLG Fire Detected

NOTE

MLG shown, EPZ/PEC operation similar.

At a preset temperature the increasing pressure within the APD reaches a point where the alarm switch closes. 28 VDC is now sent through pin A of the APD. This voltage is sent directly to the PULL FUEL/HYD OFF handle on the FPP illuminating the lights (RED) within the handle. The voltage is also sent to the control amplifier pins 34 or 35 to initiate the other associated warnings

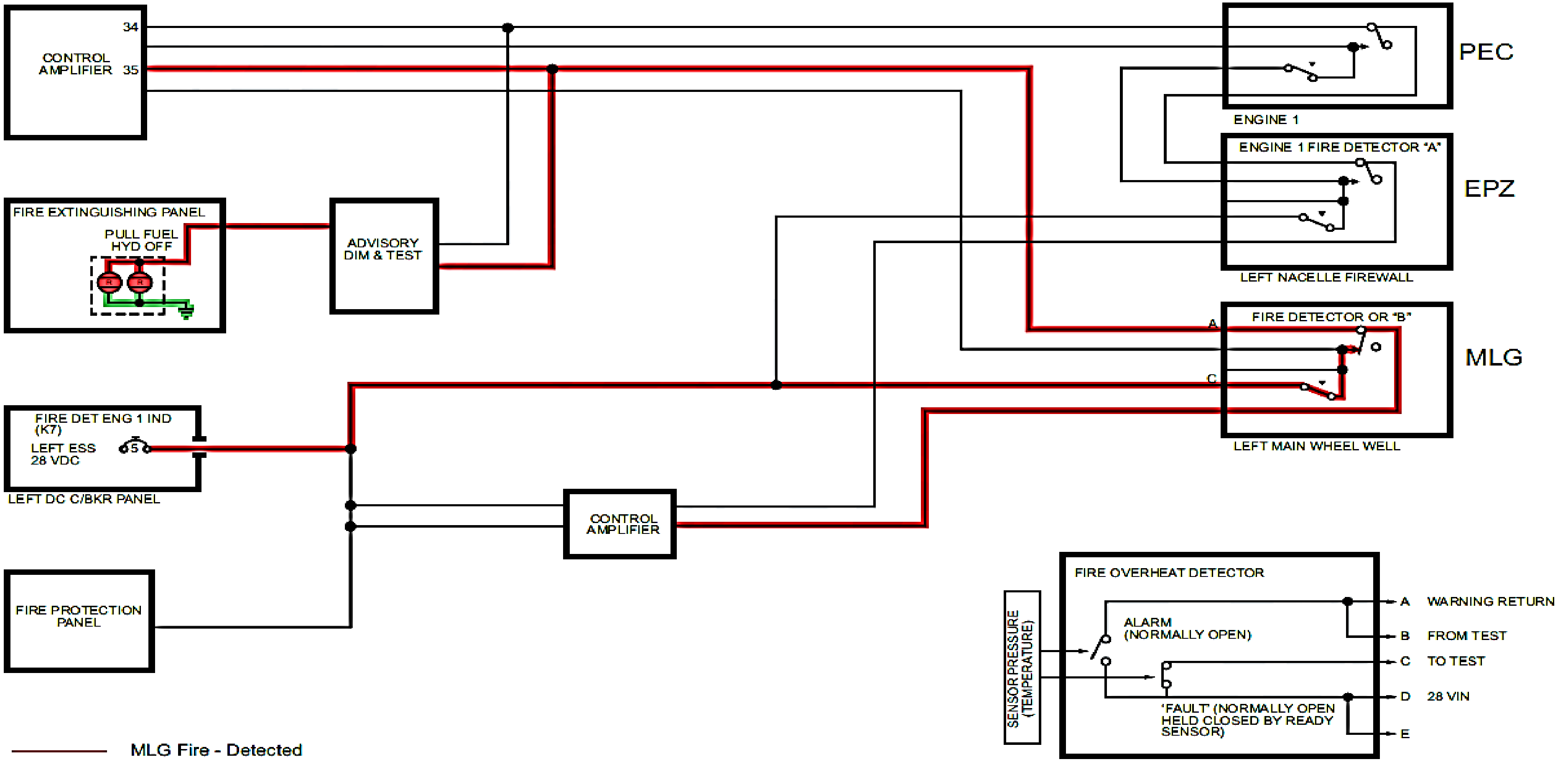


FIGURE - FIRE OVERHEAT DETECTION - MLG FIRE DETECTED

Refer Figure - PEC Fail -EPZ Detected
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The EPZ and PEC APD’s are electrically connected in such a way that it is possible to have a fault on the system and still detect a fire/overheat condition.

If PEC APD developed a leak or there was a wiring fault with the integrity circuit, the 28 VDC would not be sensed on pin 65 of the control amplifier.

The control amplifier would illuminate the Fault “A” advisory light on the FPP. If a fire/overheat condition was then sensed by EPZ, the alarm switch would close. Voltage would be sent from EPZ responder pin A to PEC responder pin B. The voltage would then be sent from PEC responder pin A directly to the PULL FUEL/ HYD OFF handle on the FPP illuminating the lights (RED) within the handle. The voltage is also sent to the control amplifier pins 34 to initiate the other associated warnings.

If EPZ APD developed a leak or there was a wiring fault with the integrity circuit, the 28 VDC would not be sensed on pin 65 of the control amplifier. The control amplifier would illuminate the Fault “A” advisory light on the FPP.

If PEC then detected a fire/overheat condition, the switch closes but there is no voltage available to signal the control amplifier.

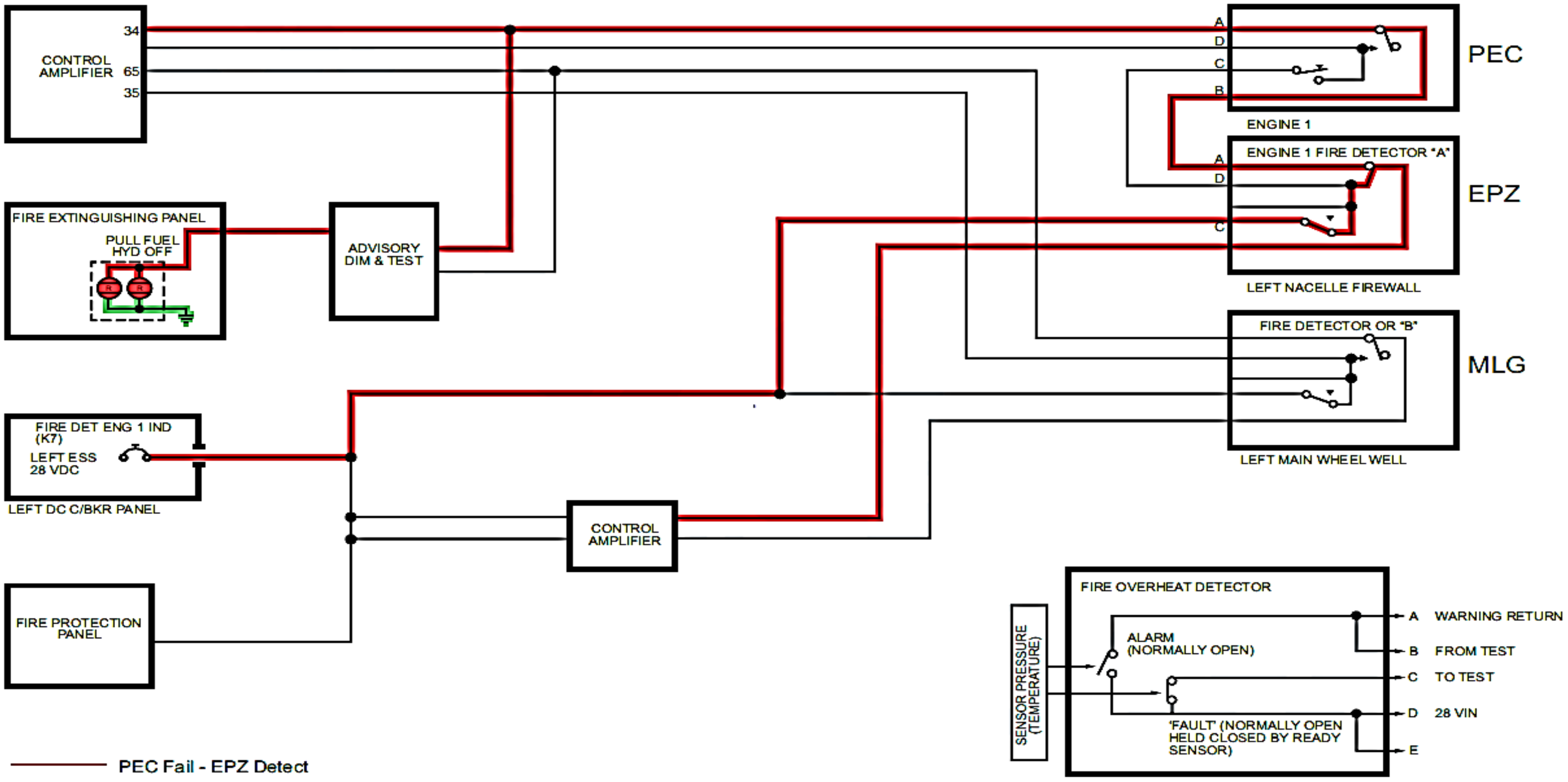


FIGURE - PEC FAIL -EPZ DETECTED

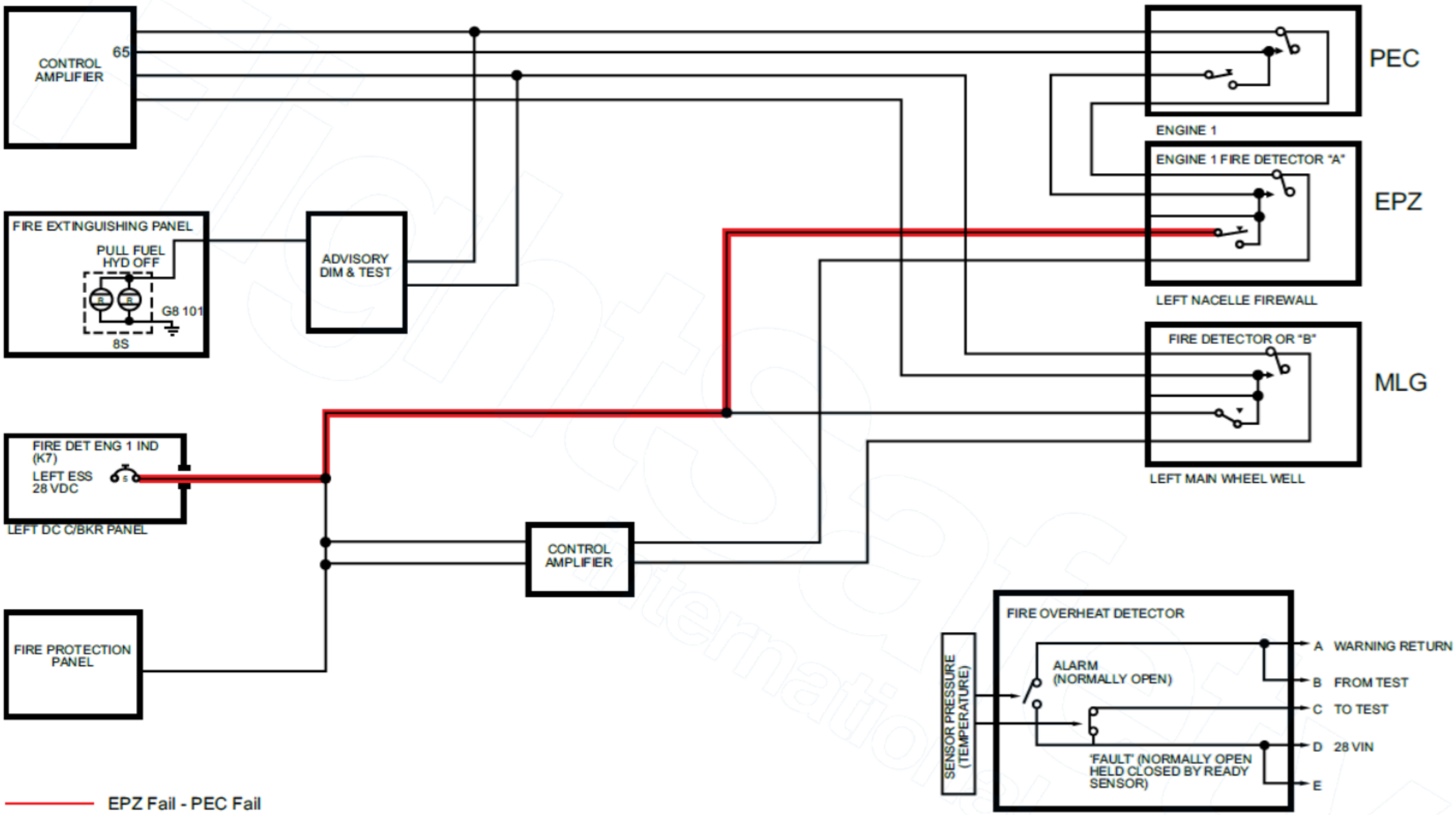


FIGURE -1 PEC FAIL - EPZ FAIL

Refer Figure - Flammable Fluid Shut-off System - Electrical Schematic

PULL FUEL/HYD OFF

The initial step for engine fire extinguishing event starts with the flammable fluid shut-off system.

For each engine the system comprises the following:

- The PULL FUEL/HYD OFF T-handle on the FPP
- Fuel and hydraulic valve position status lights on the FPP
- A motorized fuel SOV on the wing rear spar outboard of the rib at station Y112

A motorized hydraulic SOV on the right wall of the nacelle aft of the firewall (Zone 6).

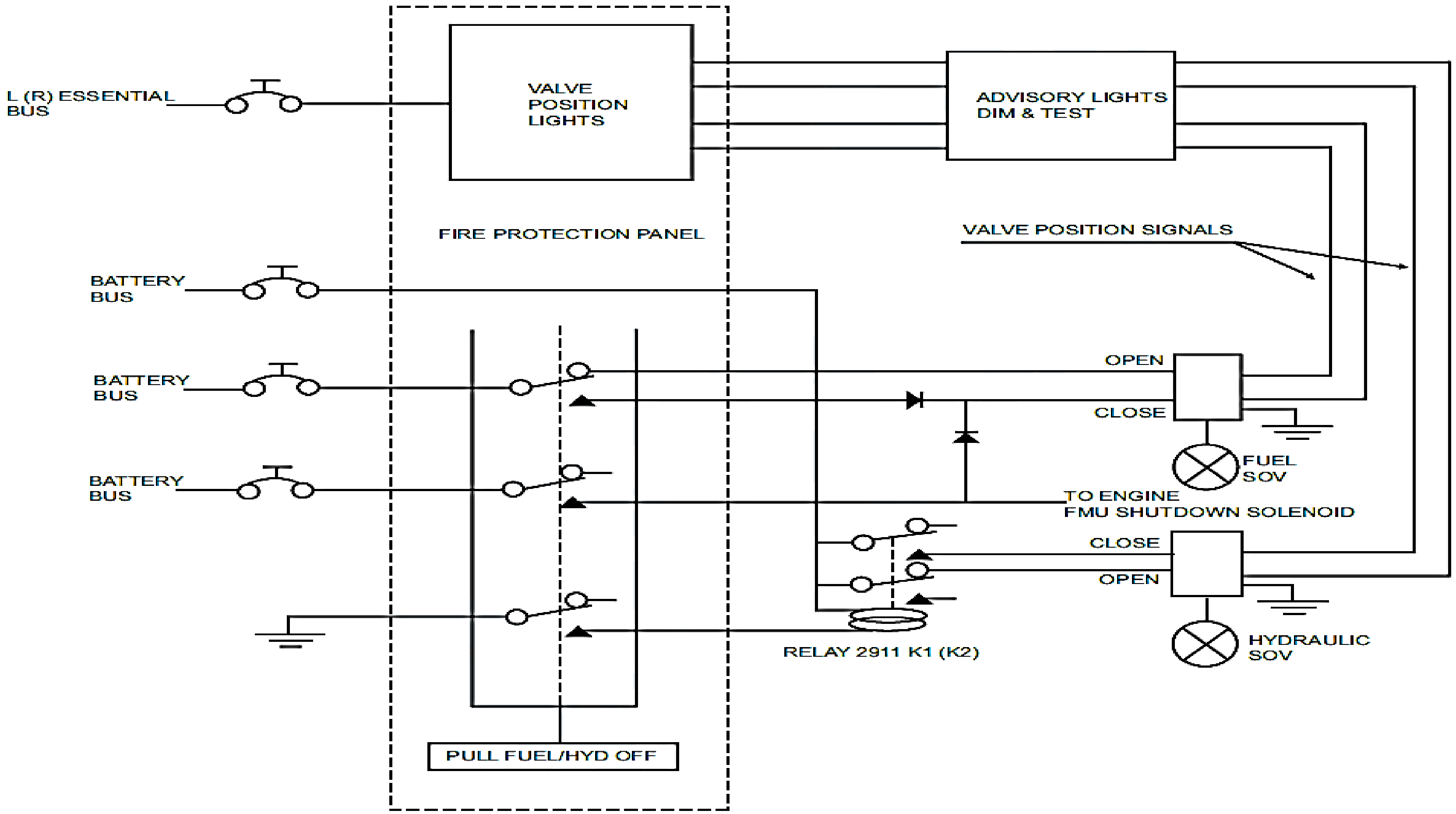


FIGURE - FLAMMABLE FLUID SHUT-OFF SYSTEM - ELECTRICAL SCHEMATIC

***Refer Figure 26- 21 Nacelle Fire Extinguishing System - Electrical Schematic (Sheet 1 of* 2)**

NOTE
Engine number 1 shown, engine number 2 operations similar.

When the PULL FUEL/HYD OFF handle is pulled, five internal contacts move from their normally closed position to the normally open position. These contacts perform the following:

- 1C - Provides a ground through pin S to hydraulic system No.1. Close the hydraulic SOV.
- 2C - Supplies 28 VDC from the battery bus to close the engine No.1 fuel SOV.
- 3C - Supplies 28 VDC from the battery bus to close the engine No.1 fuel shutoff valve. Supplies 28 VDC from the battery bus to close the airframe shut down solenoid in the No.1 engine Fuel Metering Unit (FMU).
- 4C – Supplies 28 VDC from the battery bus to the control amplifier – Signal for Bottle Arm. Supplies 28 VDC from the battery bus to the Firex switch.
- 5C – Supplies 28 VDC from the battery bus to the control amplifier – Signal for Bottle Arm. Supplies 28 VDC from the battery bus to the Firex switch.

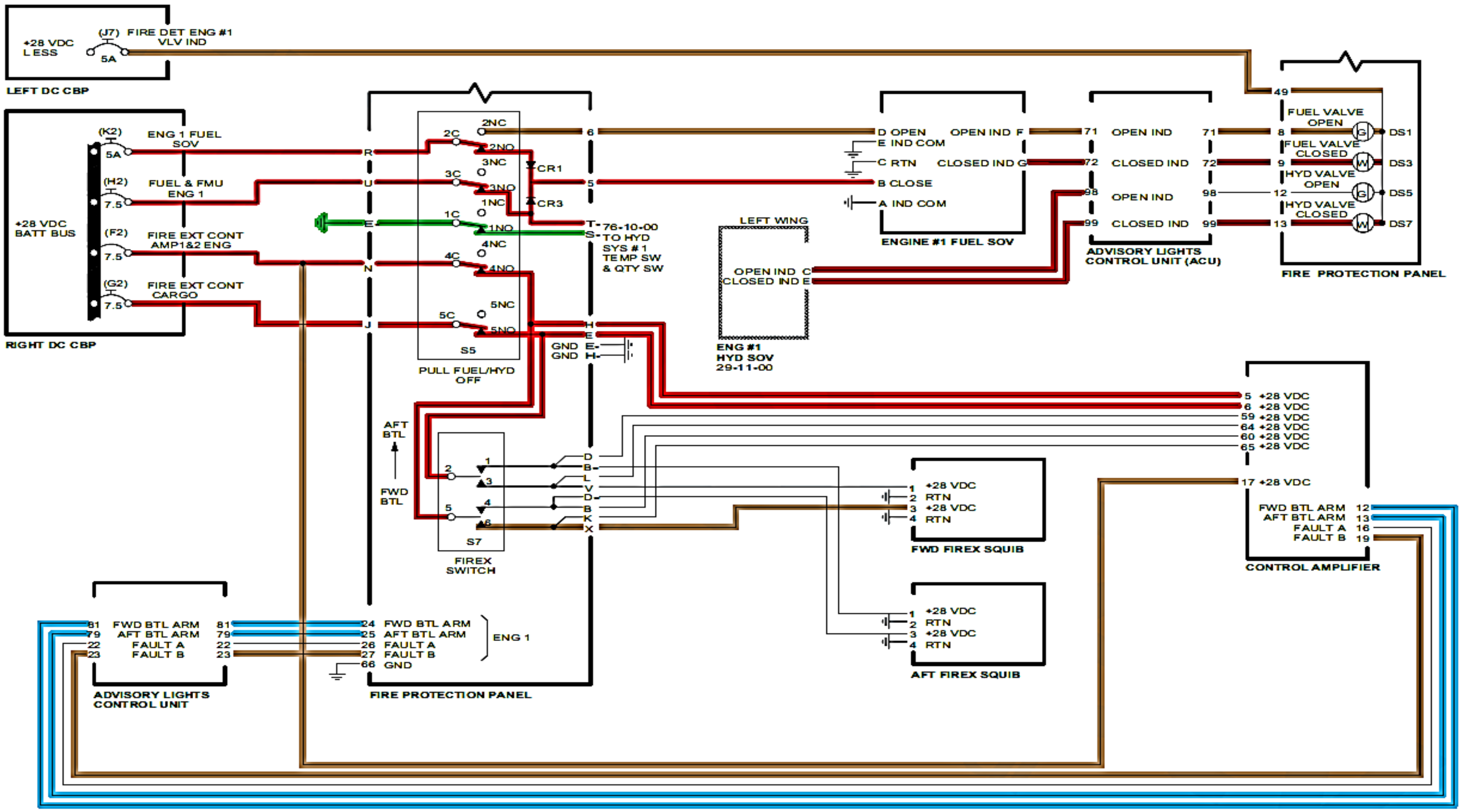


FIGURE - NACELLE FIRE EXTINGUISHING SYSTEM - ELECTRICAL SCHEMATIC (SHEET 1 OF 2)

Refer Figure 26- 22 Nacelle Fire Extinguishing System - Electrical Schematic (Sheet 2 of **2)**

NOTE
Aft Bottle Squib shown Fwd Bottle similar.

When the Firex switch is selected to AFT BTL position 28 VDC from the Batt Bus through the Fire Ext Cont Amp 1 & 2 circuit breaker is applied to pin 3 of the Aft Firex Squib. 28 VDC from the Batt Bus through Fire Ext Cont Cargo circuit breaker is applied to pin 1 of the Aft Firex Squib. This redundancy is to ensure the squib will fire releasing the extinguishing agent.

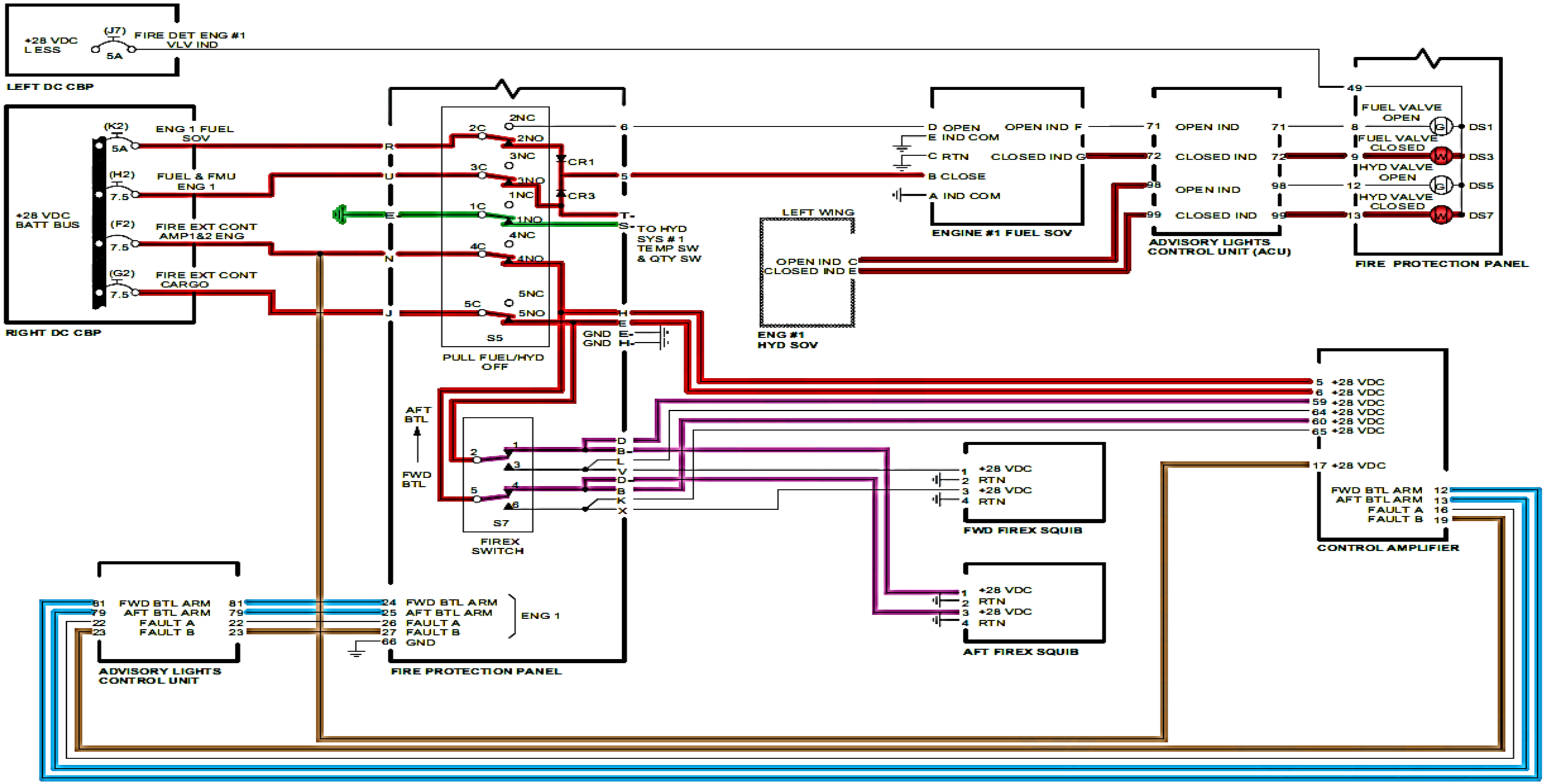


FIGURE - NACELLE FIRE EXTINGUISHING SYSTEM - ELECTRICAL SCHEMATIC (SHEET 1 OF 2)

SMOKE DETECTION SYSTEM

Introduction

The smoke detection system monitors the baggage and lavatory areas for smoke conditions.

General Description

Refer Error! Reference source not found.

Smoke detection is done by smoke detectors which are installed in the AFT baggage (cargo) compartment, the FWD baggage area and the lavatory. Smoke detectors are photosensitive devices that use a light beam that detects smoke particles when the beam is interrupted.

If a smoke or fire condition is sensed, the applicable warning indicator lights in the flight compartment, or the repeater lights in the passenger compartment ceiling come on.

The Smoke Detection System has the components that follow:

- AFT Baggage Smoke Detectors
- FWD Baggage Smoke Detector
- Lavatory Smoke Detector.

On aircraft with ModSum 4−458982, 4−458912 (extra capacity) and CR825CH03262 (cargo−combi) incorporated, the forward baggage compartment has been removed along with the smoke detector, forward HRD fire bottle, smoke switchlight and the test switch.

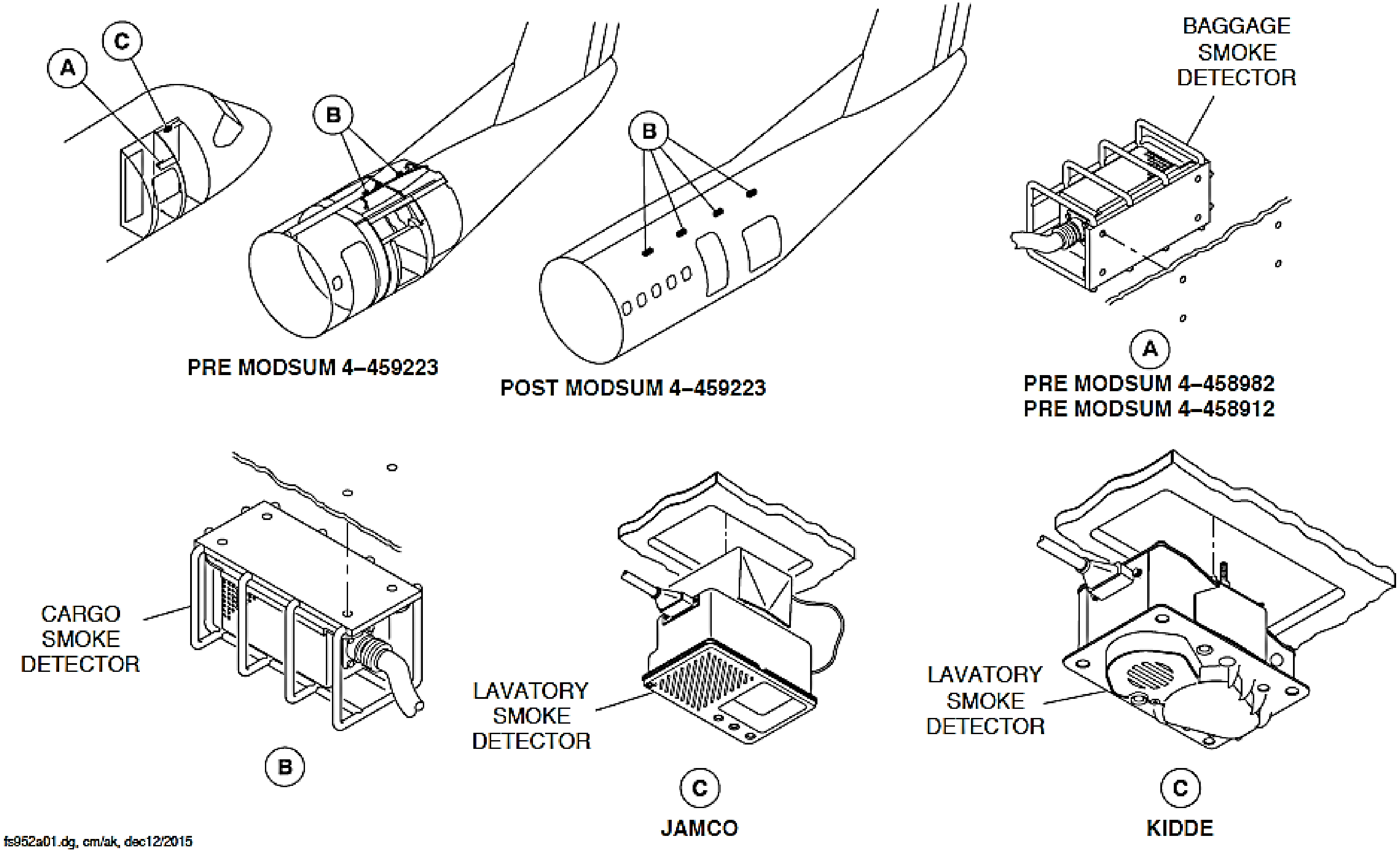


FIGURE -SMOKE DETECTORS

Detailed Description

Refer Figure -Fire Protection System Schematic

The AFT and FWD baggage smoke detectors are monitored by the control amplifier.

If a smoke or fire condition is sensed, each baggage smoke detector directly turns on the applicable SMOKE advisory light on the Fire Protection Panel (FPP). The alarm signal is also sent to the control amplifier. The control amplifier turns on indications on the FPP, Glareshield Panel, and Caution and Warning Panel in the cockpit.

Smoke in the lavatory compartment will cause the repeater lights in the passenger compartment ceiling to come on, and a single audible warning (high chime) in the passenger address system will sound. The alarm signal will switch on the Light Emitting Diode (LED) and the audio alert, located on the smoke detector. There is no cockpit indication of any smoke events in the lavatory.

Refer Figure - Fire Protection Panel Detail for AFT Baggage Compartment Smoke and Fire (Sheet 1 of 2)
Refer Figure - Warning Lights for Baggage–Compartments Smoke Detection

Smoke detection in the aft baggage compartment is annunciated by:

- SMOKE (red) warning light on the Caution and Warning Panel comes on and flashes
- Master WARNING PRESS TO RESET light on the Glareshield Panel comes on and flashes.
- BAGGAGE AFT SMOKE (red) advisory light on the Fire Protection Panel comes on
- AFT FIRE BOTTLE ARM (amber) light on the Fire Protection Panel comes on
- VENT VALVE INLET and VENT VALVE OTLT CLOSED (white) lights on the Fire Protection Panel come on
- BAGGAGE AFT EXTG (white) switchlight on the Fire Protection Panel comes on
- Three chimes.

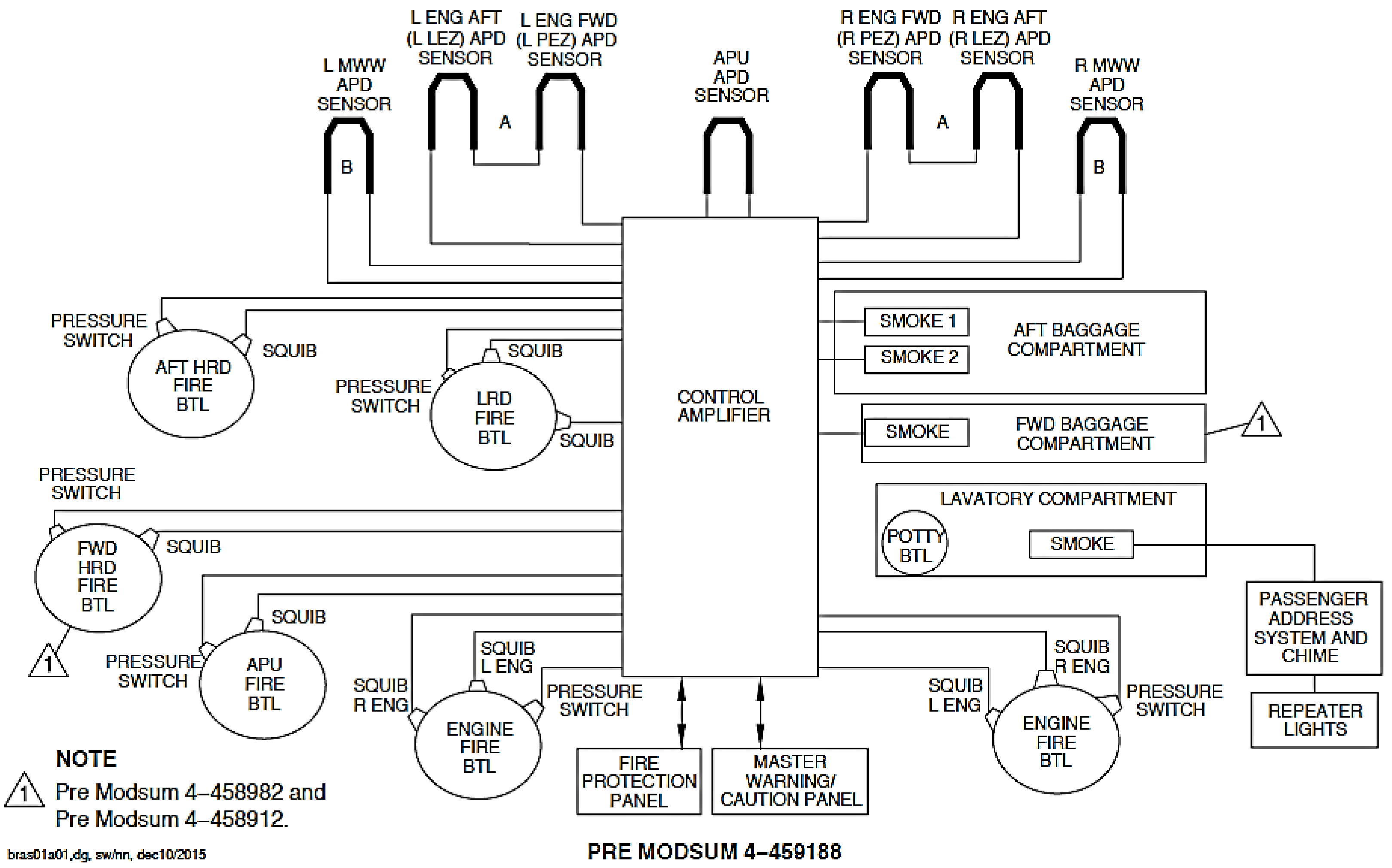


FIGURE -FIRE PROTECTION SYSTEM SCHEMATIC

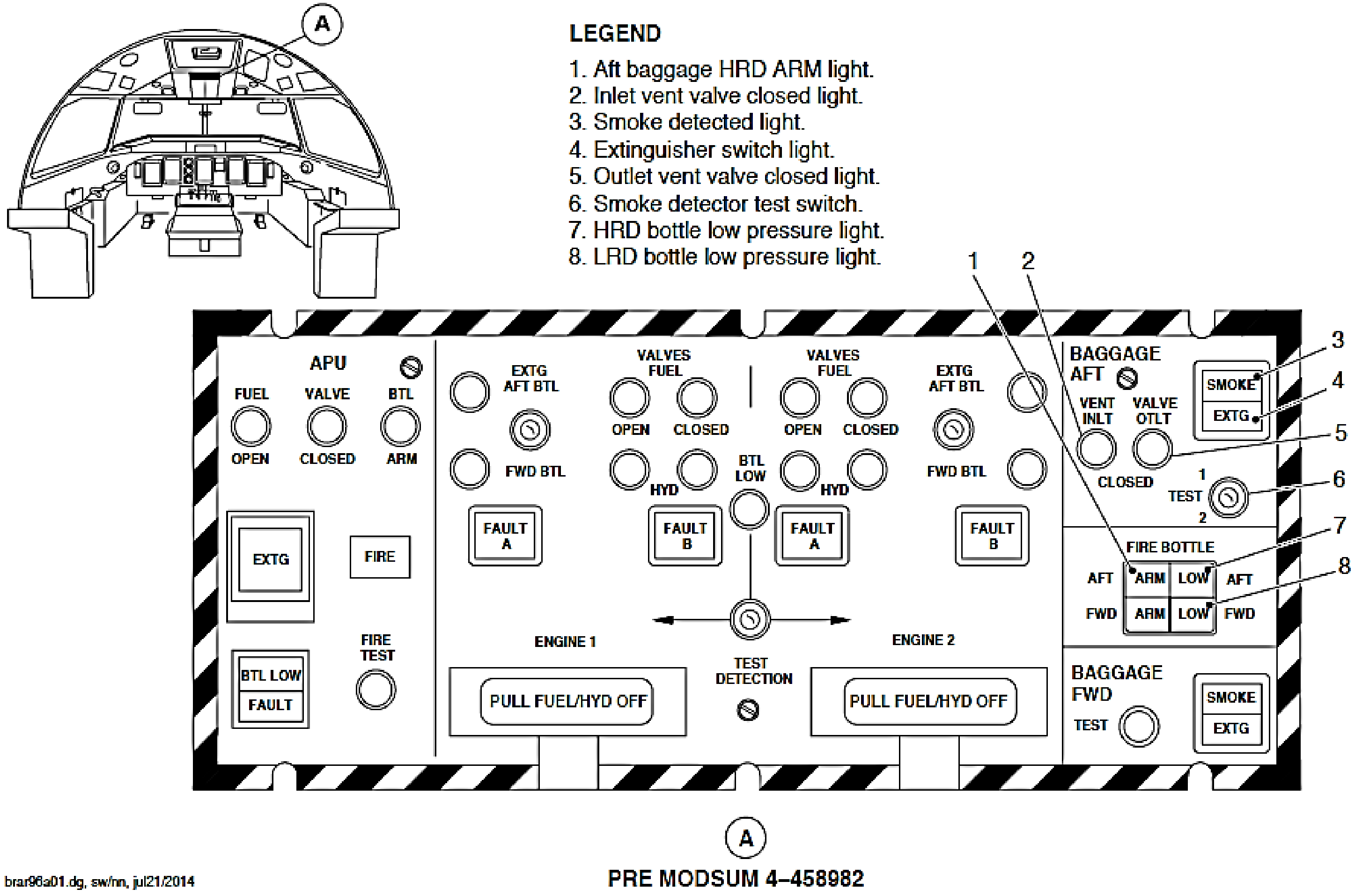
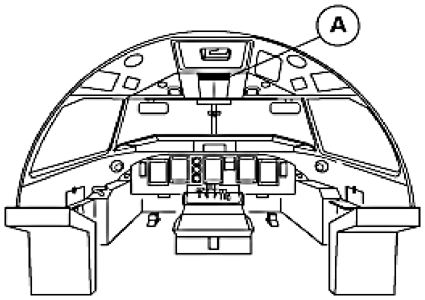
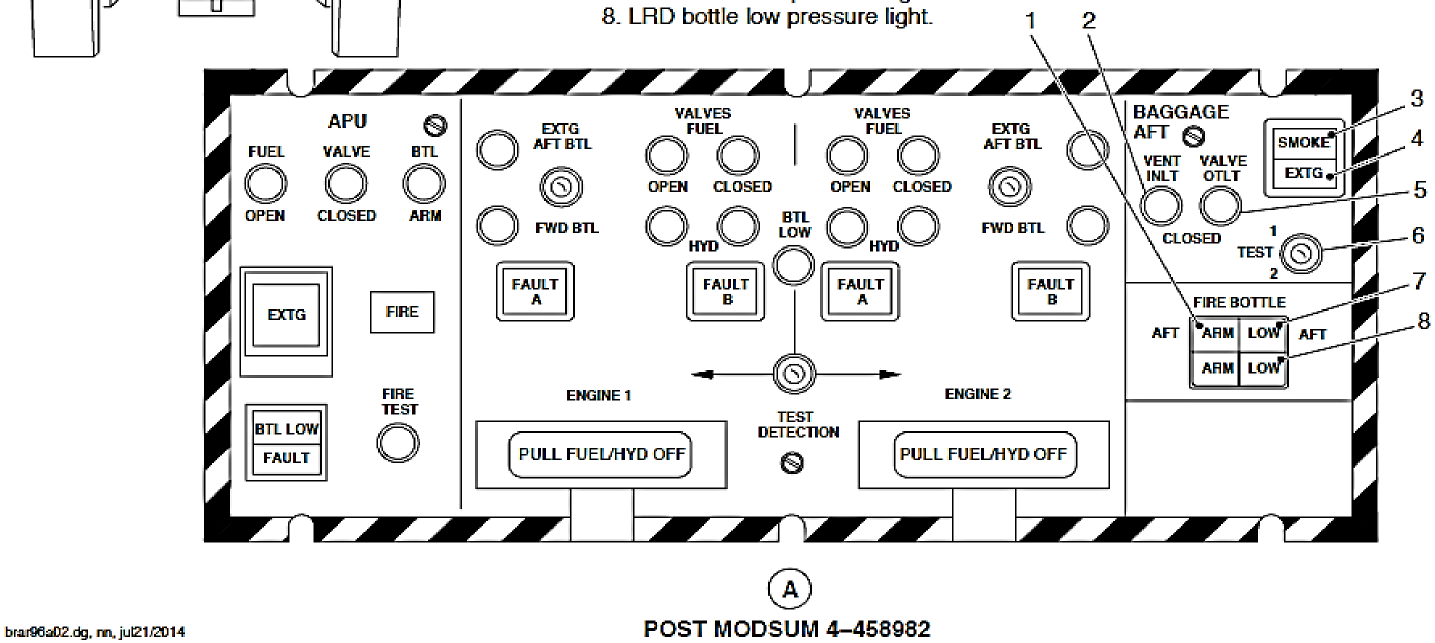


FIGURE - FIRE PROTECTION PANEL DETAIL FOR AFT BAGGAGE COMPARTMENT SMOKE AND FIRE (SHEET 1 OF 2)



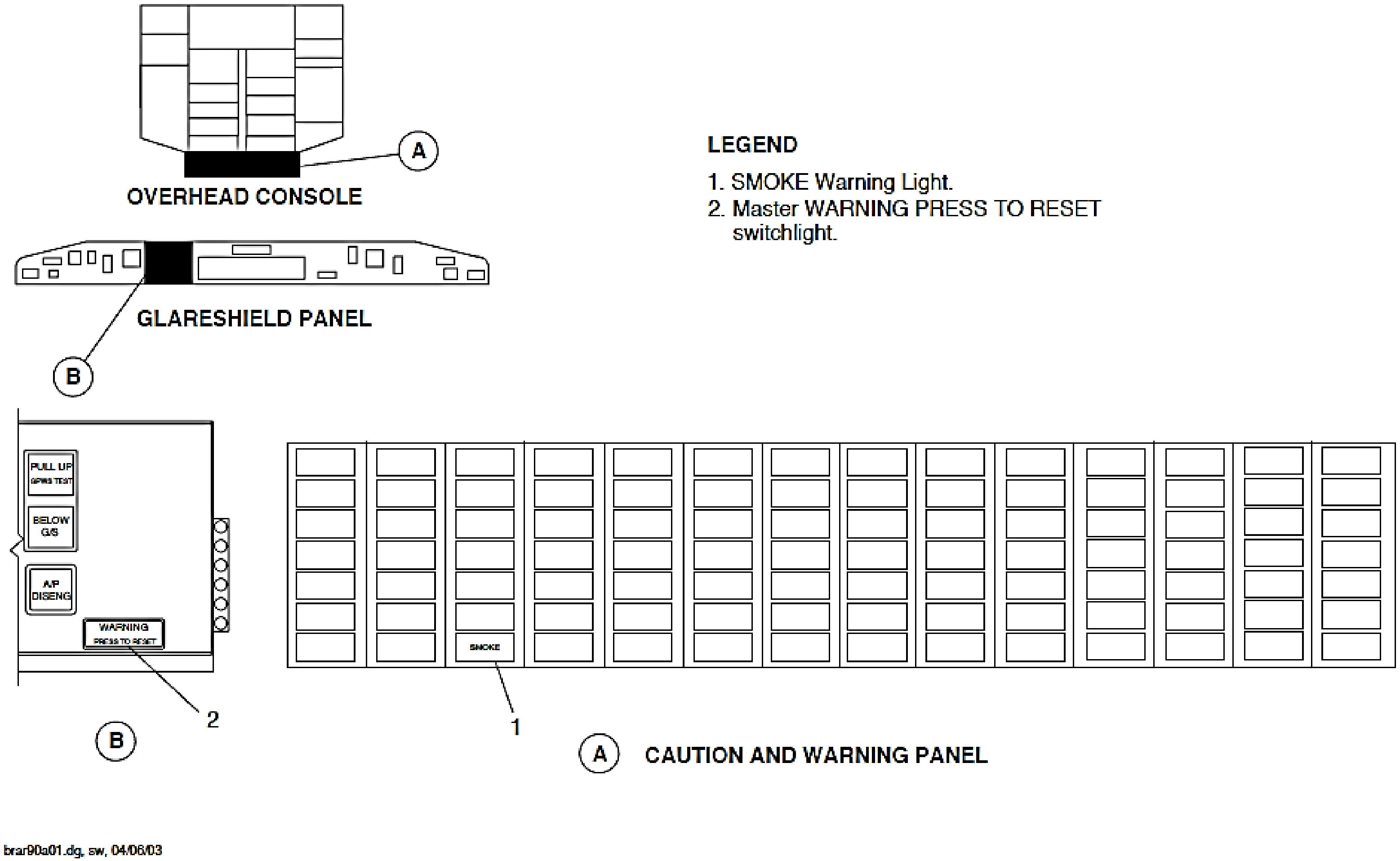
LEGEND

1. Aft baggage HRD ARM light.
2. Inlet vent valve closed light.
3. Smoke detected light.
4. Extinguisher switch light.
5. Outlet vent valve closed light.
6. Smoke detector test switch.
7. HRD bottle low pressure light.
8. LRD bottle low pressure light.



POST MODSUM 4–458982

FIGURE - FIRE PROTECTION PANEL DETAIL FOR AFT BAGGAGE COMPARTMENT SMOKE AND FIRE (SHEET 2 OF 2)



LEGEND

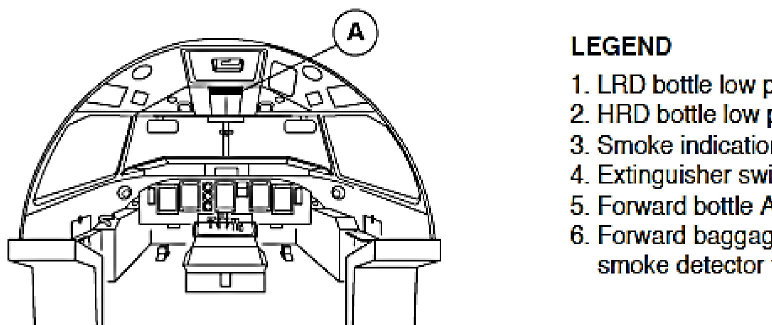
1. SMOKE Warning Light.
2. Master WARNING PRESS TO RESET switchlight.

FIGURE - WARNING LIGHTS FOR BAGGAGE−COMPARTMENTS SMOKE DETECTION

Refer Figure 26- 29 Fire Protection Panel Detail for FWD and AFT Baggage Compartment **Smoke and Fire**

Smoke detection in the forward baggage compartment is annunciated by:

- SMOKE (red) warning light on the Caution and Warning Panel comes on and flashes
- Master WARNING PRESS TO RESET light on the Glareshield Panel comes on and flashes
- BAGGAGE FWD SMOKE (red) advisory light on the Fire Protection Panel comes on
- FWD FIRE BOTTLE ARM (amber) light on the Fire Protection Panel comes on
- BAGGAGE FWD EXTG (white) switchlight on the Fire Protection Panel comes on
- Three chimes.



AFT Baggage Smoke Detectors

The AFT Baggage Smoke Detectors sense the presence of smoke in the aft baggage (cargo) compartment.

The AFT baggage compartment has two smoke detectors. Smoke detector 2 is located at the rear and smoke detector 1 is located at the front of the aft baggage compartment. Each smoke detector is a self−contained open−area detector and is a line replaceable unit (LRU). Electrical connections are provided using an external 66 pin connector, keyed to eliminate mismatching.

During a fire or smoke condition, one or both smoke detectors produce an alarm, based on a predetermined trip set point. This causes the “SMOKE” indicator light on the Fire Protection Panel (FPP) to come on. The control amplifier will automatically operate relay contacts to remove electrical power from the AFT baggage vent inlet and vent outlet shutoff valves, to minimize the air flow within the baggage compartment. Indication of vent valve status will be shown as VENT VALVE INLT and VENT VALVE OTLT lights on the FPP.

The smoke detectors have a Built−In Test capability that completely tests their functionality in response to a test signal from the FPP.

FWD Baggage Smoke Detector

The FWD Baggage Smoke Detector senses the presence of smoke in the forward baggage compartment.

The FWD baggage smoke detector is a self−contained open−area detector. It is a line replaceable unit (LRU) and is located in the forward baggage compartment. Electrical connections are provided using an external 66 pin connector, keyed to eliminate mismatching.

During a fire or smoke condition, the smoke detector produces an alarm, based on a predetermined trip set point. This causes the “SMOKE” indicator light on the Fire Protection Panel (FPP) to come on.

The smoke detector has a Built−In Test capability that completely tests its functionality in response to a test signal from the FPP.

Lavatory Smoke Detector

Refer Figure - Lavatory Smoke Detector

The Lavatory Smoke Detector senses the presence of smoke in the lavatory.

The smoke detector is a self−contained open−area detector. It is a line replaceable unit (LRU) and it is installed on the lavatory ceiling. Electrical connection is made through a D−style connector.

The aircraft is installed with a Jamco smoke detector or a Kidde Model 3000 smoke detector.

If the smoke detector is activated by a smoke condition, the lavatory smoke detector gives these indications:

- Red alarm indicator light on the smoke detector on aircraft with the Jamco smoke detector
- Red status indicator light on the smoke detector on aircraft with the Kidde smoke detector
- Local audible alarm
- One loud chime alert in the passenger address (P/A) system.

In addition to the above indications, on aircraft without CR825SO90286 (aft lavatory) and without CR825CH02917 (class divider aisle curtain), the cabin annunciator lights on all the three advisory light panels on the passenger compartment ceiling come on.

On aircraft with CR825SO90286 and with CR825CH02917, the red light on the side of the fwd lavatory comes on and the red lights on the cabin aisle class divider RHS forward come on.

On aircraft with CR825SO90286 and without CR825CH02917, the cabin annunciator lights on all the three advisory light panels on the passenger compartment ceiling come on and the red light on the side of the fwd lavatory comes on.

On aircraft with CR825SO90286, the cabin annunciator lights on all the three advisory light panels on the passenger compartment ceiling come on and the red light on the side of the aft lavatory comes on. On aircraft with the Jamco smoke detector, if the lavatory smoke alarm is activated by smoke you can cancel the alarms and reset the smoke detector when you push the interrupt push−button.

On aircraft with the Kidde smoke detector, if the lavatory smoke alarm is activated by smoke you can cancel the alarms and reset the smoke detector when you push the horn cancel push−button.

On aircraft with the Kidde smoke detector, the status indicator conditions are as follows:

- Off − smoke detector deenergized and not operational
- Steady green − normal operation
- Steady red − unserviceable (acceptable at initial power−up but unserviceable if it stays red)
- Steady red − indicates an alarm condition (smoke detected)
- Flashes green − serviceable (Type 1 fault detected)
- Flashes red − unserviceable (Type 2 fault detected)

The lavatory smoke detector is tested when you push and hold the self-test push−button switch on the detector itself (Refer to System Tests).

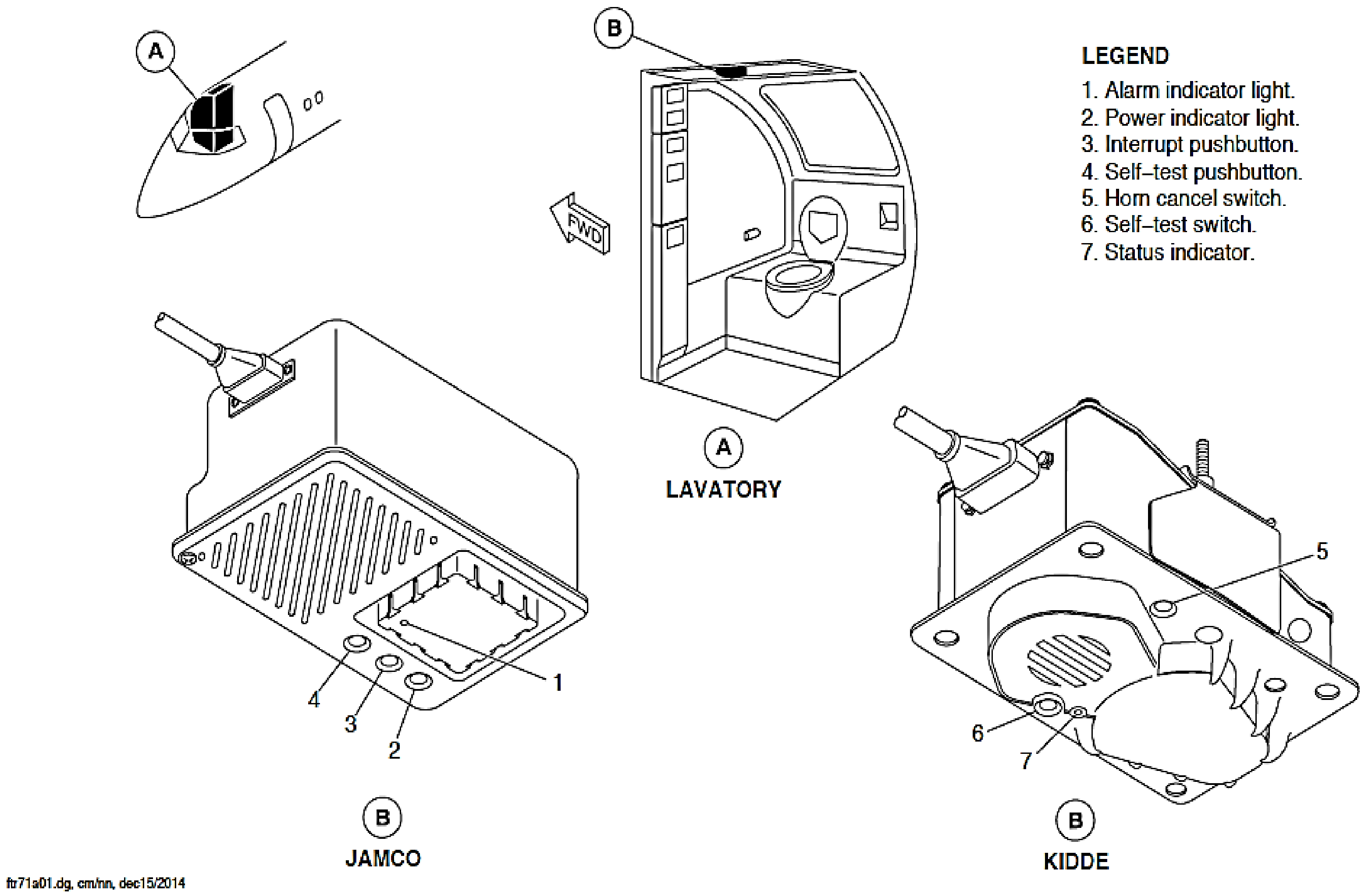


FIGURE - LAVATORY SMOKE DETECTOR

System Tests

The baggage compartment smoke test produces exactly the same response as a smoke detection event.

The AFT baggage compartment test switch is a two−position toggle switch (TEST 1/TEST 2). During TEST 1, the smoke detector 1 in the AFT baggage is tested. During TEST 2, the smoke detector 2 in the AFT baggage is tested.

During TEST 1/TEST 2 the following is annunciated:

- SMOKE (red) warning light on the Caution and Warning Panel comes on
- Master WARNING PRESS TO RESET light on the Glareshield Panel comes on and flashes
- BAGGAGE AFT SMOKE (red) advisory light on the Fire Protection Panel comes on
- AFT FIRE BOTTLE ARM (amber) light on the Fire Protection Panel comes on
- VENT VALVE INLT and VENT VALVE OTLT CLOSED (white) lights on the Fire Protection Panel come on (for the aft baggage compartment only)
- BAGGAGE AFT EXTG (white) switchlight on the Fire Protection Panel comes on
- Three chimes.

The FWD baggage compartment test switch is a push button test switch. During the forward baggage compartment test the following is annunciated:

- SMOKE (red) warning light on the Caution and Warning Panel comes on
- Master WARNING PRESS TO RESET light on the Glareshield Panel comes on and flashes
- BAGGAGE FWD SMOKE (red) advisory light on the Fire Protection Panel comes on
- FIRE BOTTLE FWD ARM (amber) light on the Fire Protection Panel comes on
- BAGGAGE FWD EXTG (white) switchlight on the Fire Protection Panel comes on
- Three chimes

The lavatory smoke detector is tested in response to a signal from a self-test push−button switch on the detector itself. To start the lavatory smoke detector test, push and hold the self-test push−button switch. The lavatory smoke detector will make the indications that follow:

- Red alarm indicator light on the smoke detector on aircraft with the Jamco smoke detector
- Red status indicator light on the smoke detector on aircraft with the Kidde smoke detector
- Local audible alarm
- One loud chime alert in the passenger address (P/A) system.

In addition to the above indications, on aircraft without CR825SO90286 (aft lavatory) and without CR825CH02917 (class divider aisle curtain), the cabin annunciator lights on all the three advisory light panels on the passenger compartment ceiling come on.

On aircraft with CR825SO90286 and with CR825CH02917, the red light on the side of the fwd lavatory comes on and the red lights on the cabin aisle class divider RHS forward come on.

On aircraft with CR825SO90286 and without CR825CH02917, the cabin annunciator lights on all the three advisory light panels on the passenger compartment ceiling come on and the red light on the side of the fwd lavatory comes on.

On aircraft with CR825SO90286, the cabin annunciator lights on all the three advisory light panels on the passenger compartment ceiling come on and the red light on the side of the aft lavatory comes on. On aircraft with the Jamco smoke detector, while you hold the self−test push−button, push and release the interrupt pushbutton. The audible alarm stops and the cabin indications go out. When you release the self−test push−button, the red alarm indicator light on the smoke detector goes off.

On aircraft with Kidde smoke detector, when you release the self−test switch, the audible alarm stops and the cabin indications go out. The red status indicator light on the smoke detector goes green or flashes green.

NOTE

The self−test push−button switch and the horn cancel switch are in a recess. To push them, it is necessary to use a thin tool approximately 3 in. (7.6 cm) long.

BAGGAGE COMPARTMENT FIRE EXTINGUISHING

Introduction

The baggage compartment fire extinguishing system suppresses fire and overheat conditions in the forward (FWD) and AFT baggage compartments.

General Description

Fire extinguishing for the FWD and AFT baggage compartments is performed by two High Rate Discharge (HRD) fire extinguisher bottles and one Low Rate Discharge (LRD) fire extinguisher bottle. Each baggage compartment has one HRD fire extinguisher; the LRD fire extinguisher is shared between the baggage compartments.

Each HRD fire extinguisher bottle has one High Rate fire extinguisher cartridge. The LRD fire extinguisher bottle has two Low Rate fire extinguisher cartridges. Each cartridge contains an Electro−Explosive Device (EED) and discharges the contents of the bottles through the applicable fire extinguisher distribution tubes.

The baggage compartment fire extinguishing system has the components that follow:

- AFT High Rate Fire Extinguisher Bottle
- FWD High Rate Fire Extinguisher Bottle
- High Rate Fire Extinguisher Cartridges
- Low Rate Fire Extinguisher Bottle
- Low Rate Fire Extinguisher Cartridges.

On aircraft with ModSum 4−458982 and ModSum 4−458912 incorporated, the forward baggage compartment has been removed along with the forward HRD fire extinguisher bottle and cartridge.

Detailed Description

Indications on the Fire Protection Panel (FPP) are used to monitor the fire and the overheat conditions in the FWD and AFT baggage compartments. Extinguisher bottle pressure is also monitored from the FPP. Pilot action, pressing the applicable SMOKE EXTG switchlight is required to activate the High Rate fire extinguishers.

If a fire condition is sensed in the AFT baggage compartment, the aft high rate fire bottle will discharge first, followed by the low rate fire extinguisher bottle which discharges automatically after seven minutes. The seven minutes delay is to make sure that any remaining fire will be extinguished.

If a fire condition is sensed in the FWD baggage compartment, both the high rate and low rate fire extinguishers will discharge at the same time.

The extinguishing event indications of an aft baggage fire are as follows:

- FIRE BOTTLE AFT ARM segment on the Fire Protection Panel goes off (blank), showing that the HRD Electro−Explosive Device (EED) is fired
- FIRE BOTTLE AFT LOW segment on the Fire Protection Panel comes on (amber), showing that the HRD fire bottle is empty (low pressure)
- FIRE BOTTLE FWD LOW segment on the Fire Protection Panel comes on (amber) 22 minutes after pushing EXTG switch, showing that the LRD fire bottle is empty (low pressure).

The extinguishing event indications of a forward baggage fire are as follows:

- FIRE BOTTLE FWD ARM segment on the Fire Protection Panel goes off (blank), showing that the HRD EED is fired
- FIRE BOTTLE FWD LOW segment on the Fire Protection Panel comes on (amber), showing that the HRD fire bottle is empty (low pressure)
- FIRE BOTTLE AFT LOW segment on the Fire Protection Panel comes on (amber) 15 minutes after pushing EXTG switch, showing that the LRD fire bottle is empty (low pressure).

Aft High Rate Fire Extinguisher Bottle

The Aft High Rate Fire Extinguisher Bottle contains the suppressant necessary for first line fire protection.

The AFT high rate fire bottle is a stainless steel container filled with 11 lb. (4.99 kg) of liquid fire extinguishing agent Halon 1301 and pressurized with Nitrogen gas at 360 psig (2482 kPag). The bottle is installed in the aft equipment bay. One high rate fire extinguisher cartridge is located on the bottom of the AFT high rate fire bottle.

If a smoke condition is sensed in the AFT baggage compartment, the AFT inlet and outlet vent valves will close automatically and give an indication on the FPP. The HRD extinguisher fires with the push of the AFT EXTG switch. After a seven minute delay, the low rate fire extinguisher will automatically discharge to make sure the fire is out.

On A/C with modsum 4Q126352 incorporated, the electrical connectors are attached with lanyards. These lanyards make sure that the correct connector is connected to the discharge squib during maintenance practices.

Low Rate Fire Extinguisher Bottle

The Low Rate Fire Extinguisher Bottle contains the suppressant necessary for slow release.

The low rate fire extinguisher is a stainless steel container filled with 17.5 lb. (7.94 kg) of liquid fire extinguishing agent Halon 1301 and pressurized with Nitrogen gas at 360 psig (2482 kPag). The bottle is installed in the aft equipment bay, but is shared between the FWD and AFT baggage compartments.

When a smoke condition is sensed in the AFT baggage compartment and the aft High−Rate fire bottle is activated, the Low−Rate fire extinguisher will automatically discharge after seven minutes. A metering device located inside the low rate fire extinguisher controls the slow release of the extinguishing agent from the bottle. This device lets the bottle discharge the extinguishing agent into the baggage compartment at a concentration of 3% for 45 minutes.

If a smoke condition is sensed in the FWD baggage compartment, both the High−Rate and the Low−Rate fire extinguishers will discharge at the same time.

On A/C with modsum 4Q126352 incorporated, the electrical connectors are attached with lanyards. These lanyards make sure that the correct connector is connected to the discharge squib during maintenance practices.

On aircraft with ModSum 4−124265 and 4−190605 incorporated, the discharge tube is removed from the forward discharge head and blanked with a plug.

On aircraft with ModSum 4−124541 incorporated, the forward discharge head is blanked with a plug.

Dryer Metering Unit (DMU)

On aircraft with ModSum 4−459188 incorporated, the low rate fire extinguisher bottles are connected with the DMU. The moisture from the extinguisher agent is removed by the DMU to prevent freezing. The DMU is a one−time use desiccant unit that must be replaced when a low rate fire extinguisher bottle is discharged.

FWD High Rate Fire Extinguisher Bottle

The FWD High Rate Fire Extinguisher Bottle contains the suppressant necessary for first line fire protection.

There is one High Rate fire extinguisher bottle located on the forward baggage compartment aft wall. The High Rate extinguisher for the FWD baggage compartment is a stainless steel container filled with 4.33 lb (1.96 kg) of liquid fire extinguishing agent Halon 1301 and pressurized with Nitrogen gas at 360 psig (2482 kPag).

If a smoke condition is sensed in the FWD baggage compartment an indication is given on the FPP. Pilot action is required to manually activate the high rate fire extinguisher and the low rate fire extinguisher will discharge simultaneously with it.

On A/C with modsum 4Q126352 incorporated, the electrical connectors are attached with lanyards. These lanyards make sure that the correct connector is connected to the discharge squib during maintenance practices.

High Rate Fire Extinguisher Cartridges

High Rate Fire Extinguisher Cartridges cause the extinguisher bottle to discharge its contents through the distribution tubes leading to the FWD or AFT baggage compartment.

There is one High Rate fire extinguisher cartridge for each High Rate fire extinguisher. Each fire extinguisher cartridge is attached to its related discharge head, located on the bottom of the fire extinguisher bottle.

Each cartridge contains a dual bridgewire Electro−Explosive Device (EED) and redundant power lines with separate circuit breakers for mission reliability. In each cartridge, bridgewire “A” is taken from one circuit breaker and routed down the left side of the aircraft to the cartridge. Bridgewire “B” is taken from a separate circuit breaker and routed down the right side of the aircraft to the cartridge. Loss of electrical power to one bridgewire will not cause loss of extinguisher fire capability.

Each cartridge has an electrical connector. When an electrical signal is sent through the bridgewires, the EED ignites and explodes rupturing a “burst” disk. This causes the extinguisher bottle to discharge its contents through the distribution tubes leading to the FWD or AFT baggage compartment.

Low−Rate Fire Extinguisher Cartridges

Low−Rate Fire Extinguisher Cartridges cause the extinguisher bottle to discharge its contents through the distribution tubes leading to the AFT or FWD baggage compartment.

There are two low rate fire extinguisher cartridges. Each fire extinguisher cartridge is attached to its related discharge head, located on the bottom of the LRD fire extinguisher bottle.

Each cartridge contains a dual bridgewire Electro−Explosive Device (EED) and redundant power lines with separate circuit breakers for mission reliability. In each cartridge, bridgewire "A" is taken from one circuit breaker and routed down the left side of the aircraft to the cartridge. Bridgewire "B" is taken from a separate circuit breaker and routed down the right side of the aircraft to the cartridge. Loss of either power breaker will not cause loss of fire extinguisher capability.

Each cartridge has a related electrical connector. When an electrical signal is sent through the bridgewires, the EED ignites and explodes rupturing a burst disk. This causes the extinguisher bottle to discharge its contents through the distribution tubes leading to the AFT or FWD baggage compartment.

On aircraft with ModSums 4−458927, 4−124541 and 4−190605 incorporated, there is only one low rate fire extinguisher cartridge, installed at the aft discharge head for the aft baggage compartment. The forward discharge head is blanked with a plug.

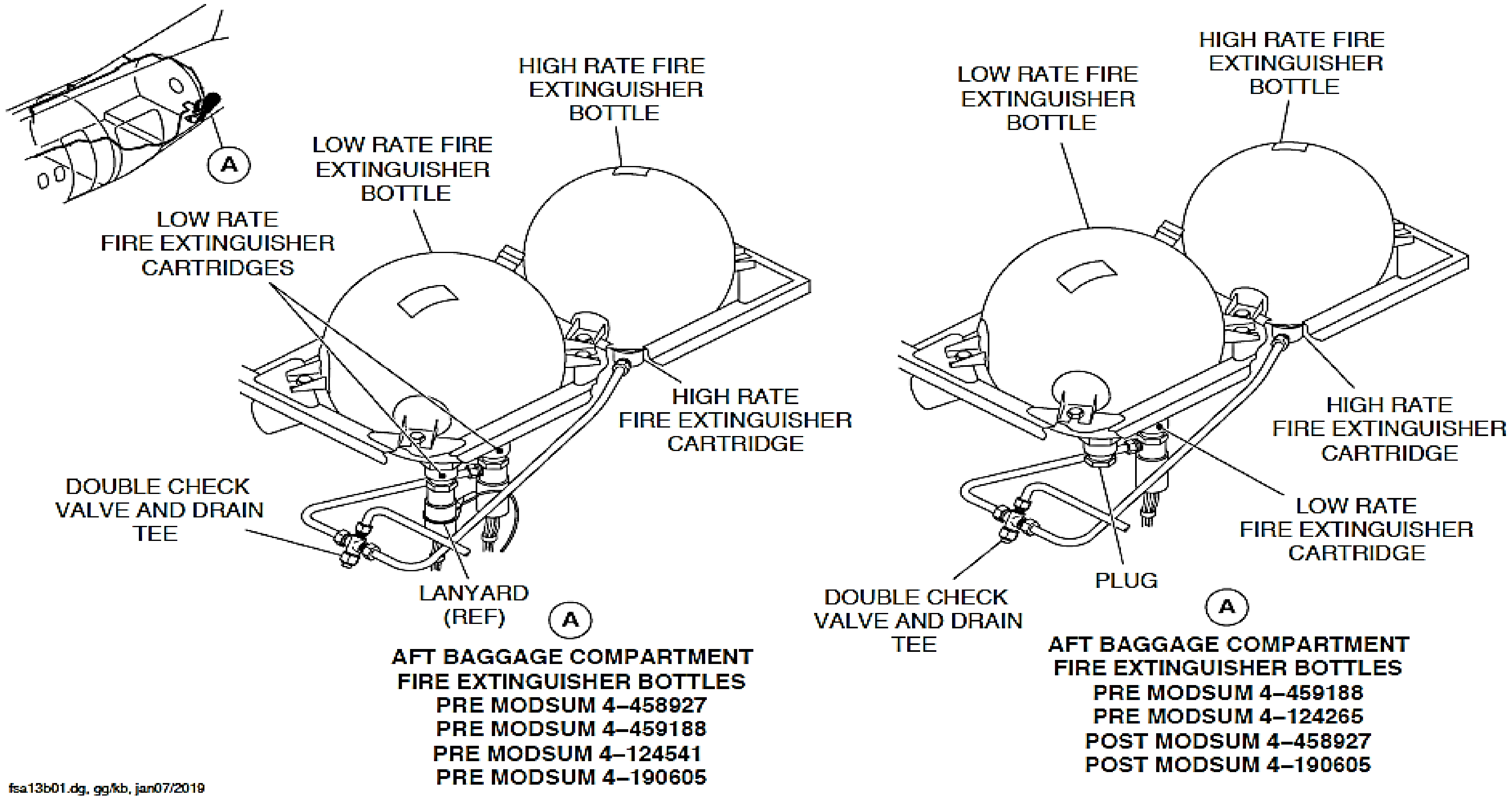


FIGURE - AFT BAGGAGE COMPARTMENT FIRE EXTINGUISHER

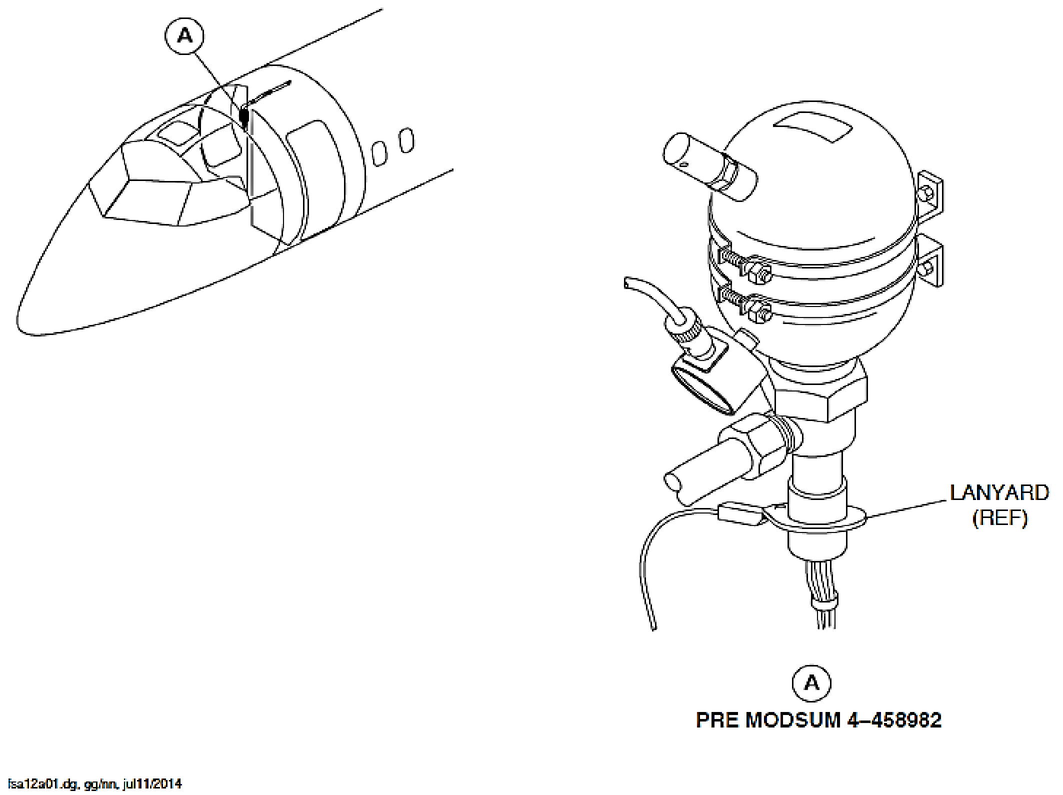


FIGURE - FORWARD BAGGAGE COMPARTMENT FIRE EXTINGUISHER

Refer Figure - Smoke Detected

The FWD baggage smoke detector and the AFT baggage smoke detector 2 (Aft) receive 28 VDC from the right essential bus via circuit breaker M5, SMK DET BAG/CGO 2. The AFT baggage smoke detector 1 (Fwd) and the LAV smoke detector receive 28 VDC from the right essential bus via circuit breaker L5, SMK DET LAV/CGO.

Smoke detected by the FWD baggage smoke detector outputs 28 VDC on pin D directly to the FPP to illuminate the SMOKE (Red) advisory switchlight. This signal is also sent to the fire control amplifier pin 8. The control amplifier outputs to illuminate the following:

- The SMOKE red switchlight
- The EXT white switchlight
- The FWD ARM amber light and
- The SMOKE warning light & Master WARNING.

Smoke detected by either AFT baggage smoke detector outputs 28 VDC on pin D directly to the FPP to illuminate the SMOKE (Red) advisory switchlight.

This signal is also sent to the fire control amplifier pin 9. The control amplifier outputs to close the inlet and the outlet valve and to illuminate the following:

- The SMOKE red switchlight
- The EXT white switchlight
- The AFT ARM amber light
- The SMOKE warning light & Master WARNING.

The smoke detection system will automatically arm the appropriate squibs if the squibs have bridgewire continuity and power. The smoke detection system will reset upon removal of the smoke. However, the inlet and outlet valves in the AFT baggage compartment will remain closed until the appropriate circuit breakers are reset.

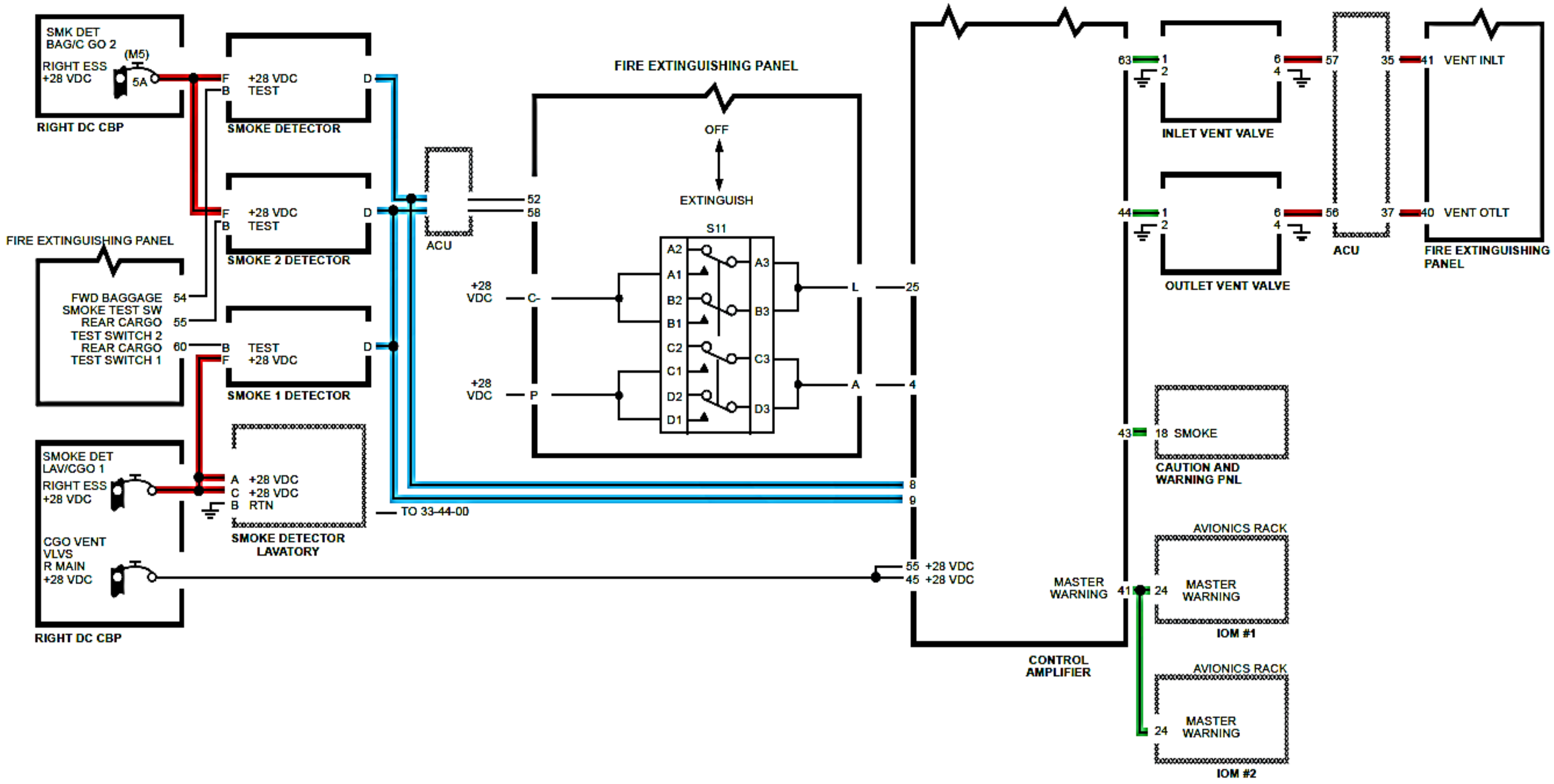


FIGURE - SMOKE DETECTED

Refer Figure - Aft Baggage Smoke Detected

Pressure in the AFT HRD bottle and LRD bottle is monitored by the control amplifier. The pressure switches receive 28 VDC from the left essential bus via the Cargo LTS circuit breaker K3.

The continuity of the squibs is also monitored. If both the HRD bottle pressure switch shows pressure and the squib shows continuity when smoke is detected, the control amplifier will output a ground to light the appropriate ARM light.

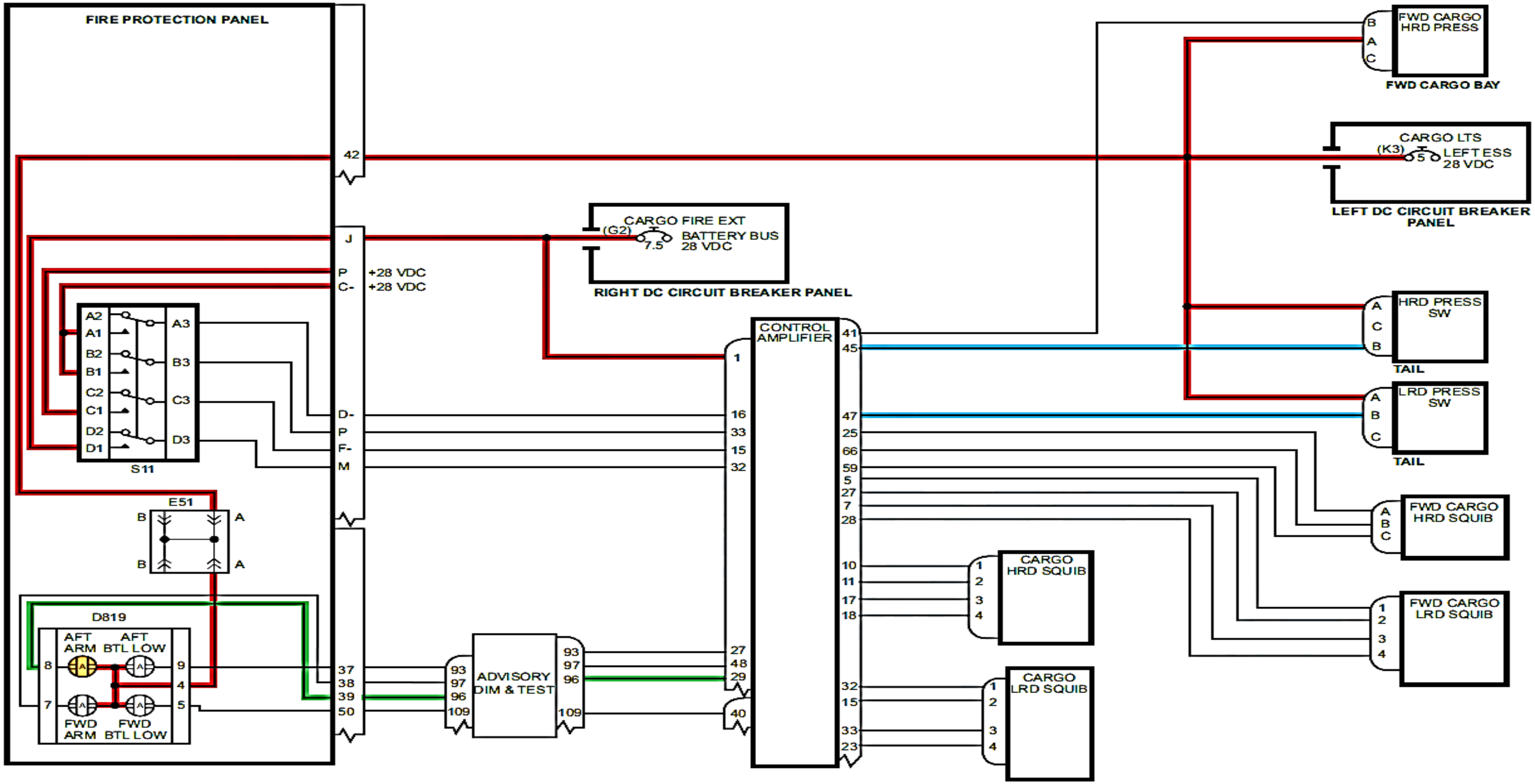


FIGURE - AFT BAGGAGE SMOKE DETECTED

Refer Figure - Aft Baggage Fire Extinguished (Sheet 1 of 2)

When the extinguisher switch is pushed, 28 VDC is supplied to the control amplifier pins 15, 16, 32 and 33.

The control amplifier outputs 28 VDC to the AFT HRD bottle on 2 separate lines and the cartridge fires. The HRD bottle pressure goes low, therefore, the armed light extinguishes and the BTL LOW light illuminates.

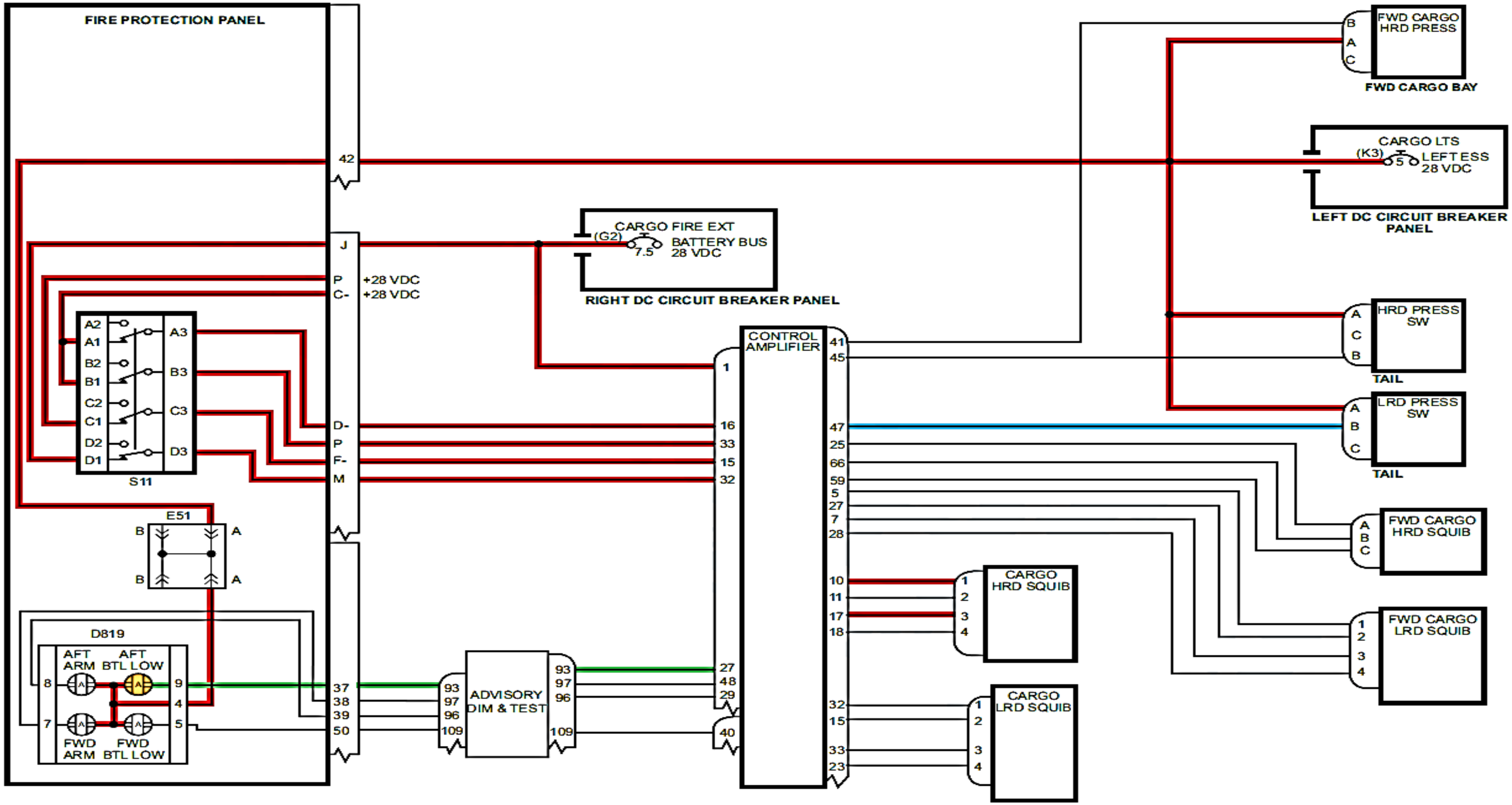


FIGURE - AFT BAGGAGE FIRE EXTINGUISHED (SHEET 1 OF 2)

Refer Figure - Aft Baggage Fire Extinguished (Sheet 2 of 2)

Seven minutes after the extinguish switch is pushed, the control amplifier outputs 28 VDC to the AFT cargo squib on the LRD bottle. Due to flow restriction, it takes 15 minutes before the LRD pressure switch will show low pressure. At this time the FWD BTL LOW advisory light will illuminate.

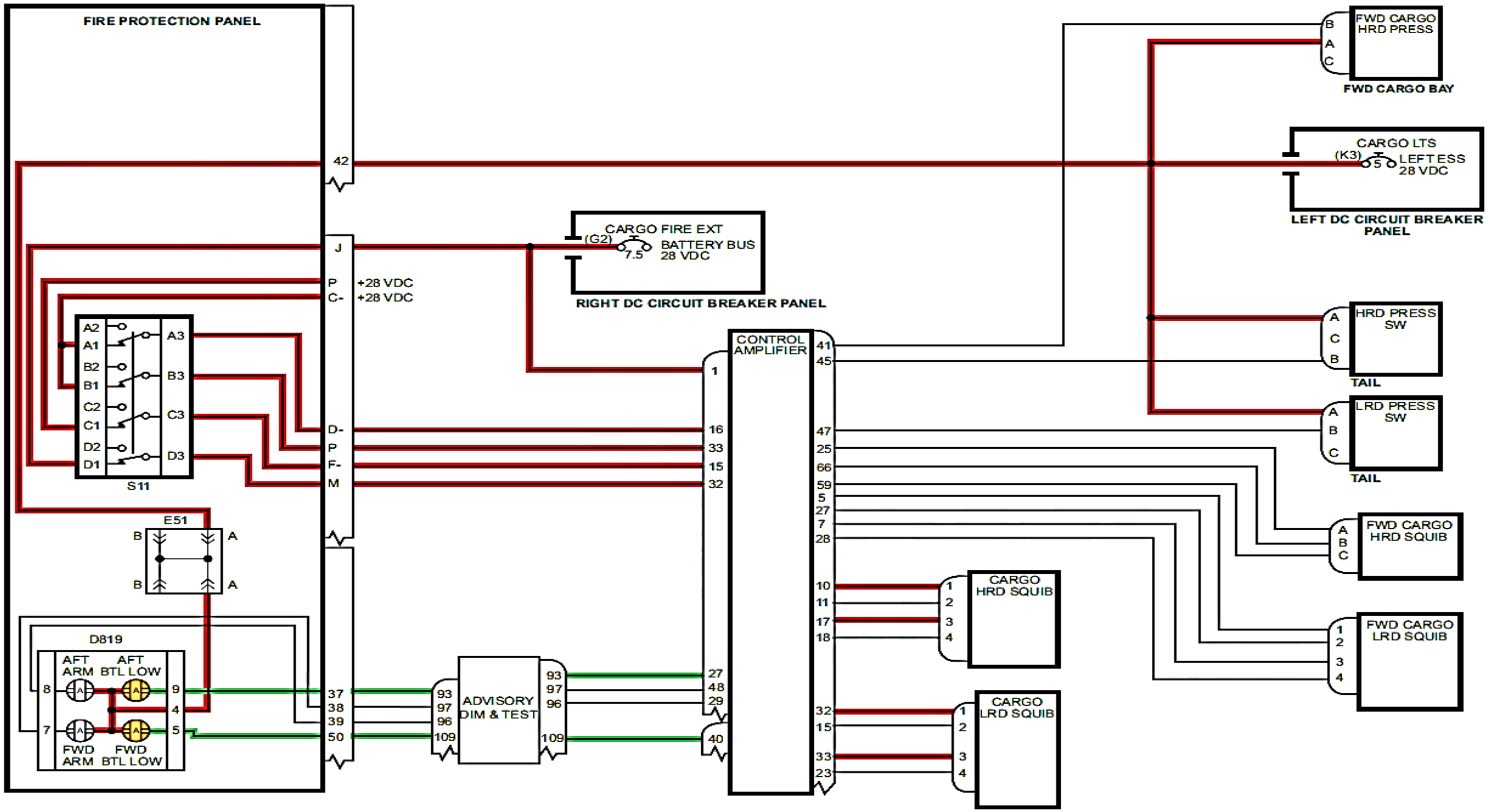


FIGURE - AFT BAGGAGE FIRE EXTINGUISHED (SHEET 2 OF 2)

Refer Figure - Forward Baggage Compartment Fire Detected

Pressure in the FWD HRD bottle and LRD bottle is monitored by the control amplifier. The pressure switches receive 28 VDC from the left essential bus via the Cargo LTS circuit breaker K3.

The continuity of the squibs is also monitored. If both the FWD bottle pressure switch shows pressure and the squib shows continuity when smoke is detected, the control amplifier will output a ground to light the appropriate ARM light.

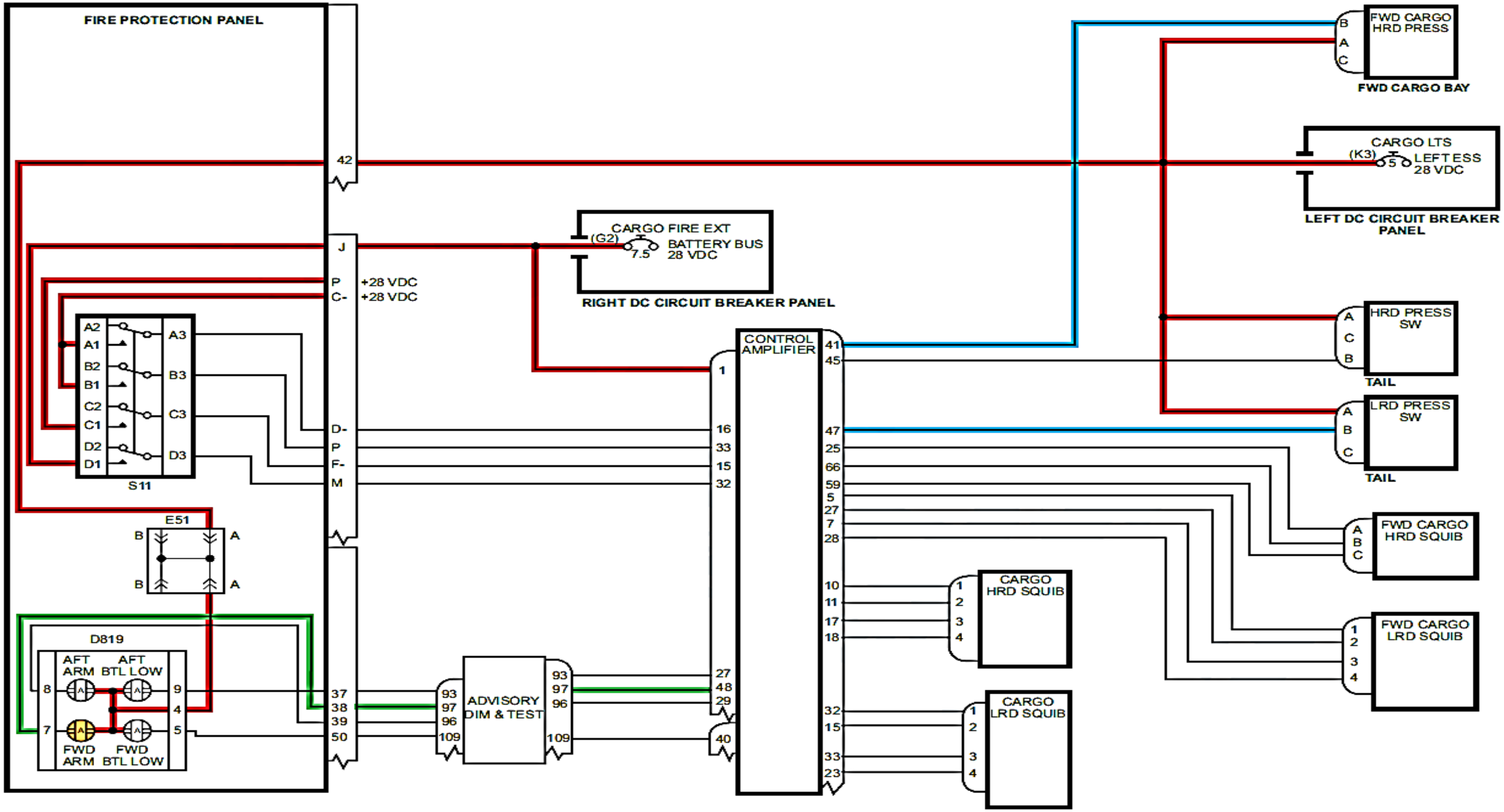


FIGURE - FORWARD BAGGAGE COMPARTMENT FIRE DETECTED

Refer Figure - Forward Baggage Compartment Fire Extinguished (Sheet 1 of 2)

When the extinguisher switch is pushed, 28 VDC is supplied to the control amplifier pins 15, 16, 32 and 33.

The control amplifier outputs 28 VDC to the FWD HRD and the LRD bottles on 2 independent lines and the cartridges fire. The HRD bottle pressure goes low, therefore, the armed light extinguishes and the BTL LOW light illuminates.

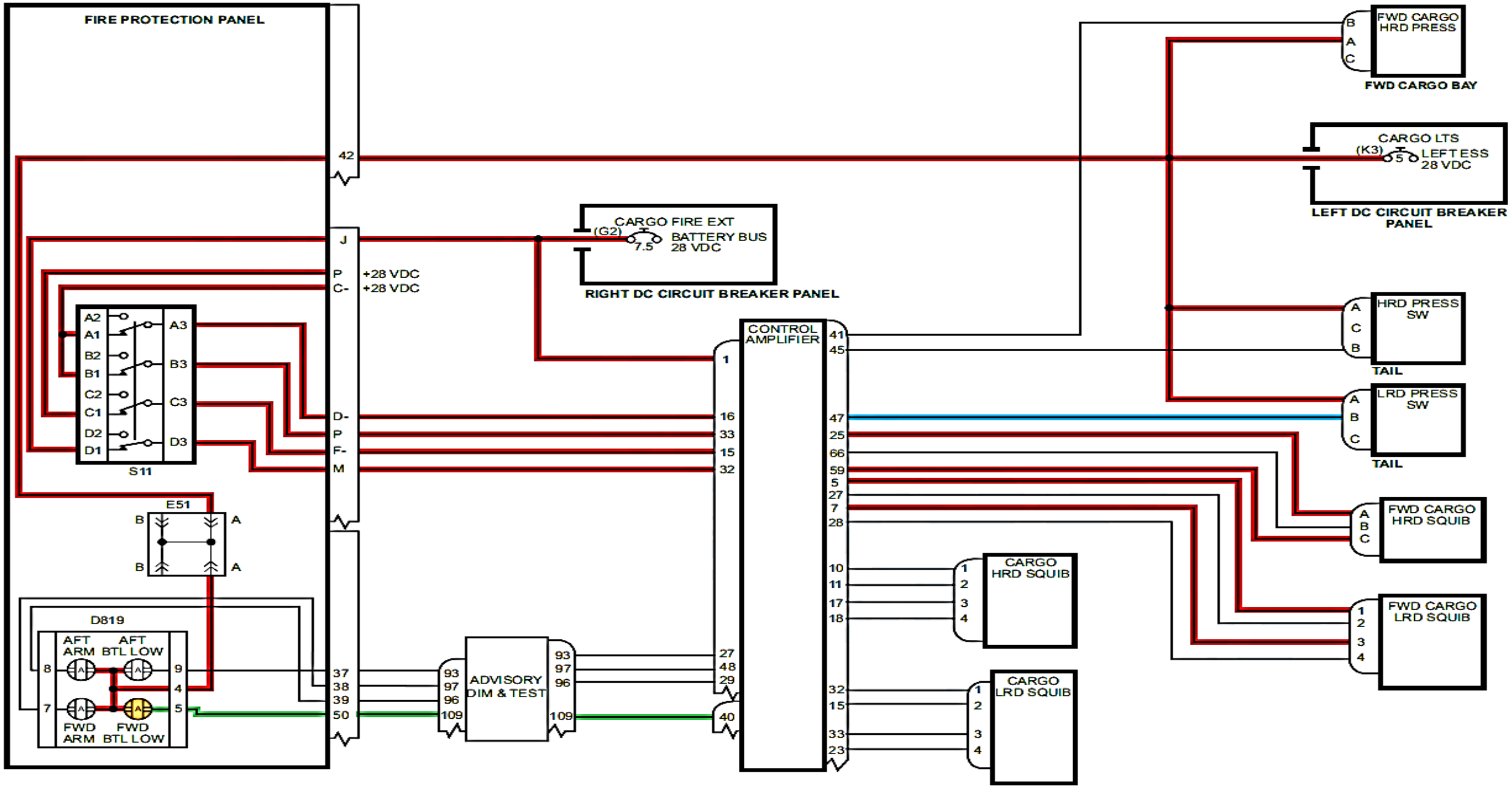


FIGURE - FORWARD BAGGAGE COMPARTMENT FIRE EXTINGUISHED (SHEET 1 OF 2)

Refer Figure - Forward Baggage Compartment Fire Extinguish (Sheet 2 of 2)

The HRD and LRD bottles fire at the same time. Due to restrictors, the LRD bottle pressure switch will not activate until 15 minutes after squib discharge. At this time the AFT BTL LOW advisory light illuminates.

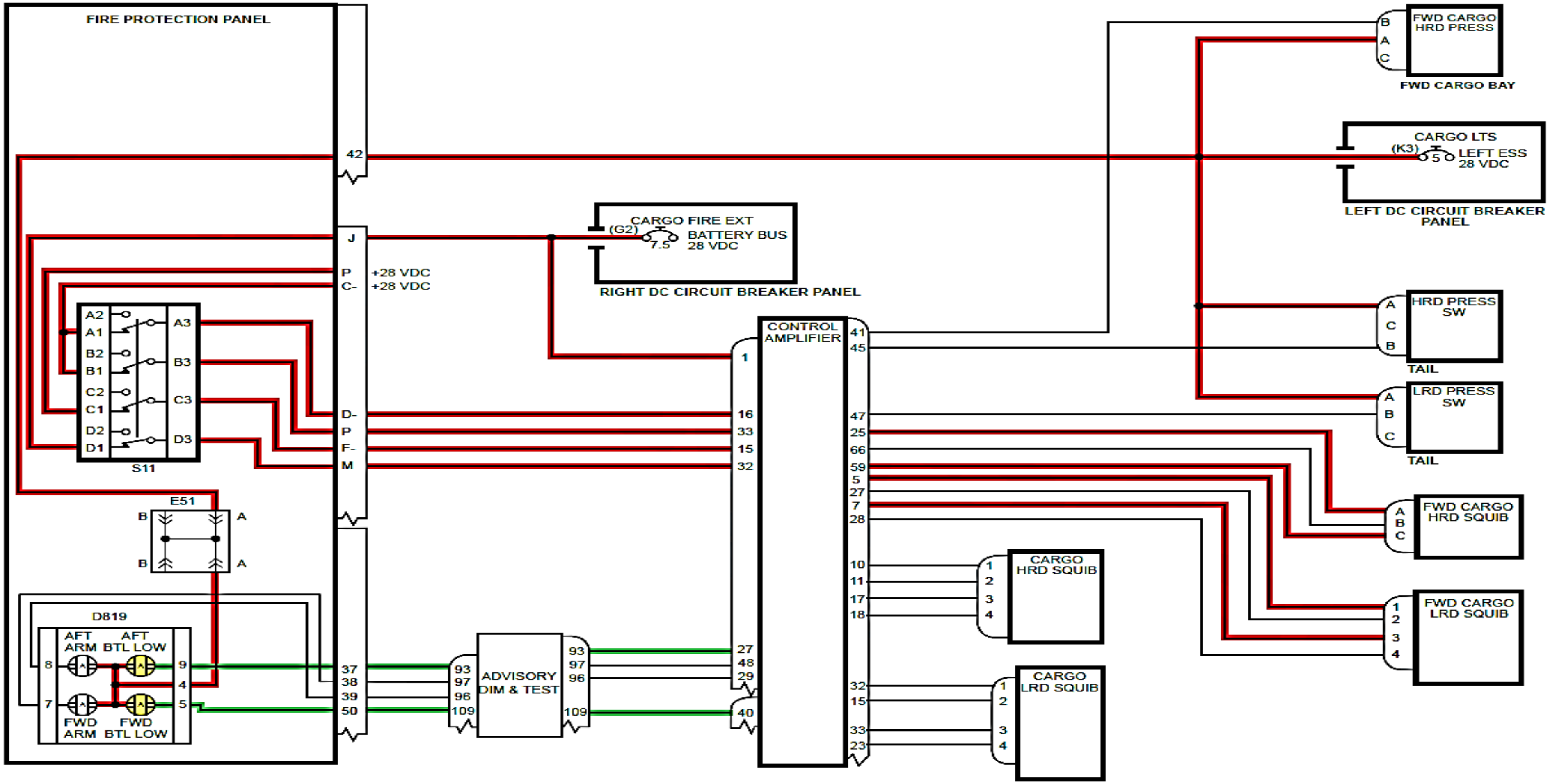


FIGURE - FORWARD BAGGAGE COMPARTMENT FIRE EXTINGUISH (SHEET 2 OF 2)

APU FIRE DETECTION SYSTEM

Introduction

The Auxiliary Power Unit (APU) fire detection system senses fire or overheat conditions in the APU compartment.

General Description

The APU fire detection system has one Advanced Pneumatic Detector (APD) located in the APU compartment which senses changes in temperature.

The APU fire detection system has the component that follows:

- APU Compartment Advanced Pneumatic Detector.

Detailed Description

Refer Figure - Advanced Pneumatic Detector for APU Compartment Fire Detection

There is one Advanced Pneumatic Detector (APD) located in the APU compartment. The APD has a main element body and a sensor tube which extends around the APU compartment. The main element body is held in place by two P−clamps and the sensor tube is supported by additional clamps and grommets. The sensor tube is filled with an inert gas which is sensitive to changes in temperature.

Refer Figure - Advanced Pneumatic Detector (APD)

The APD receives electrical power from the right essential bus. Two switches (integrity and alarm) are integral in the APD. They show the APU compartment fire or overheat conditions on the Fire Protection Panel (FPP).

The integrity switch monitors the pressure in the sensor element. If an APD breaks, the loss of pressure will open the integrity switch and signal the control amplifier to turn on the FAULT light on the FPP.

The alarm switch is normally open, and closes when an overheat or fire condition occurs, caused by the pressure increase in the APD. The table that follows shows the specified alarm temperature values:

ADVANCED PNEUMATIC DETECTORS (APD) LOCATION	DETECTOR PART NUMBER	ALARM TEMPERATURE °F (°C)	MINIMUM RESET TEMPERATURE °F (°C)
APU Zone	8SC1289	564 to 636 (296 to 336)	504 (262)

The alarm switch directly signals the FIRE advisory light on the FPP. The FIRE (red) advisory light remains on steady state for the duration of the alarm. The alarm signals are also processed by the control amplifier and then sent to the FPP.

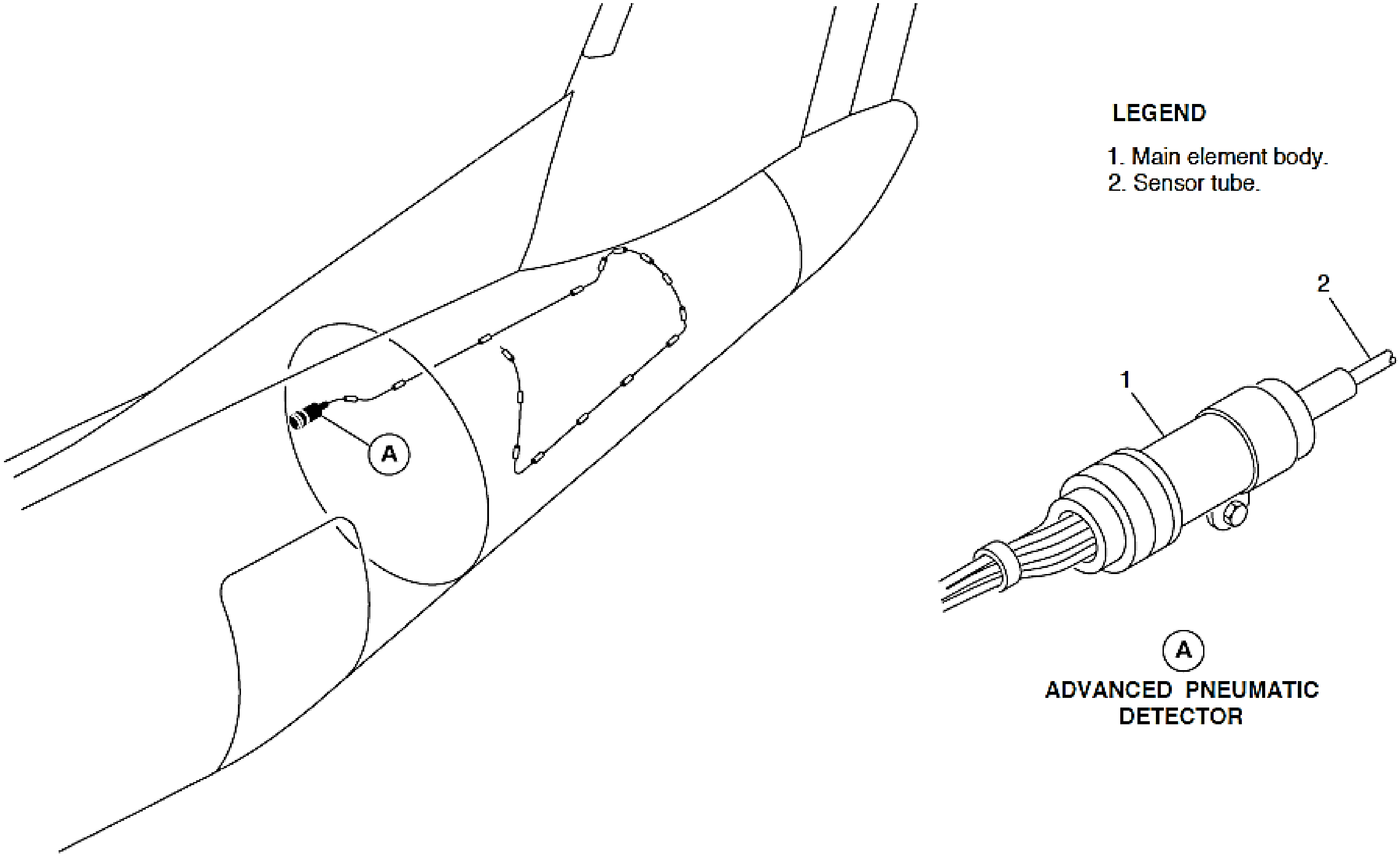


FIGURE - ADVANCED PNEUMATIC DETECTOR FOR APU COMPARTMENT FIRE DETECTION

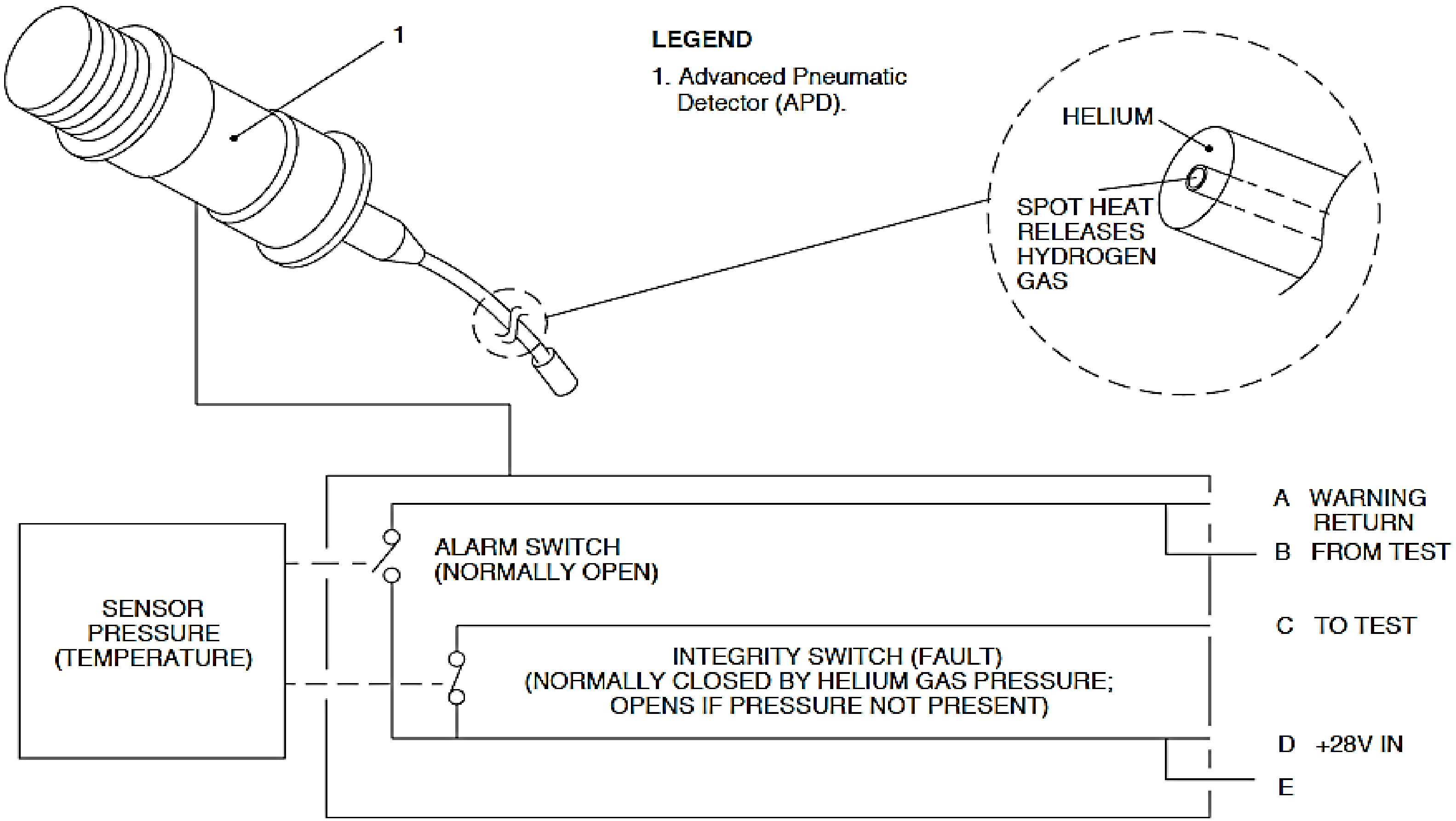


FIGURE - ADVANCED PNEUMATIC DETECTOR (APD)

Refer Figure - Fire Protection Panel Detail for APU Fire Protection
Refer Figure - Indications for APU Fire

When a fire or overheat is sensed in the APU, the control amplifier does as follows:

- FIRE (red) light on the Fire Protection Panel (FPP) comes on
- BTL ARM (amber) light on the FPP comes on
- FUEL OPEN (green) advisory light on the FPP goes off
- FUEL CLOSED (white) light on the FPP comes on
- EXTG (white) switch light on the FPP comes on
- CHECK FIRE DET (red) warning light on the Caution and
- Warning Panel comes on and flashes
- Master WARNING PRESS TO RESET (red) light on the Glareshield Panel comes on and flashes
- Fire Bell (optional).

When a fire or overheat is sensed, the APU will automatically shut down and the FAIL advisory light on the APU control panel will come on.

System Tests

To test the APU Fire Protection System, the FIRE TEST push button on the Fire Protection Panel (FPP) is pushed. During APU fire test, the control amplifier performs the following:

- FIRE (red) light on the Fire Protection Panel (FPP) comes on
- BTL ARM (amber) light on the FPP comes on
- FUEL OPEN (green) advisory light on the FPP goes off
- FUEL CLOSED (white) light on the FPP comes on
- EXTG (white) switch light on the FPP comes on
- FAULT (amber) light on the FPP comes on

- CHECK FIRE DET (red) warning light on the Caution and Warning Panel comes on and flashes
- Master WARNING PRESS TO RESET (red) light on the Glareshield Panel comes on and flashes
- Fire Bell (optional).

The shutdown signal to the APU FADEC and the APU fuel shut off valve circuits are also tested and the APU shuts down.

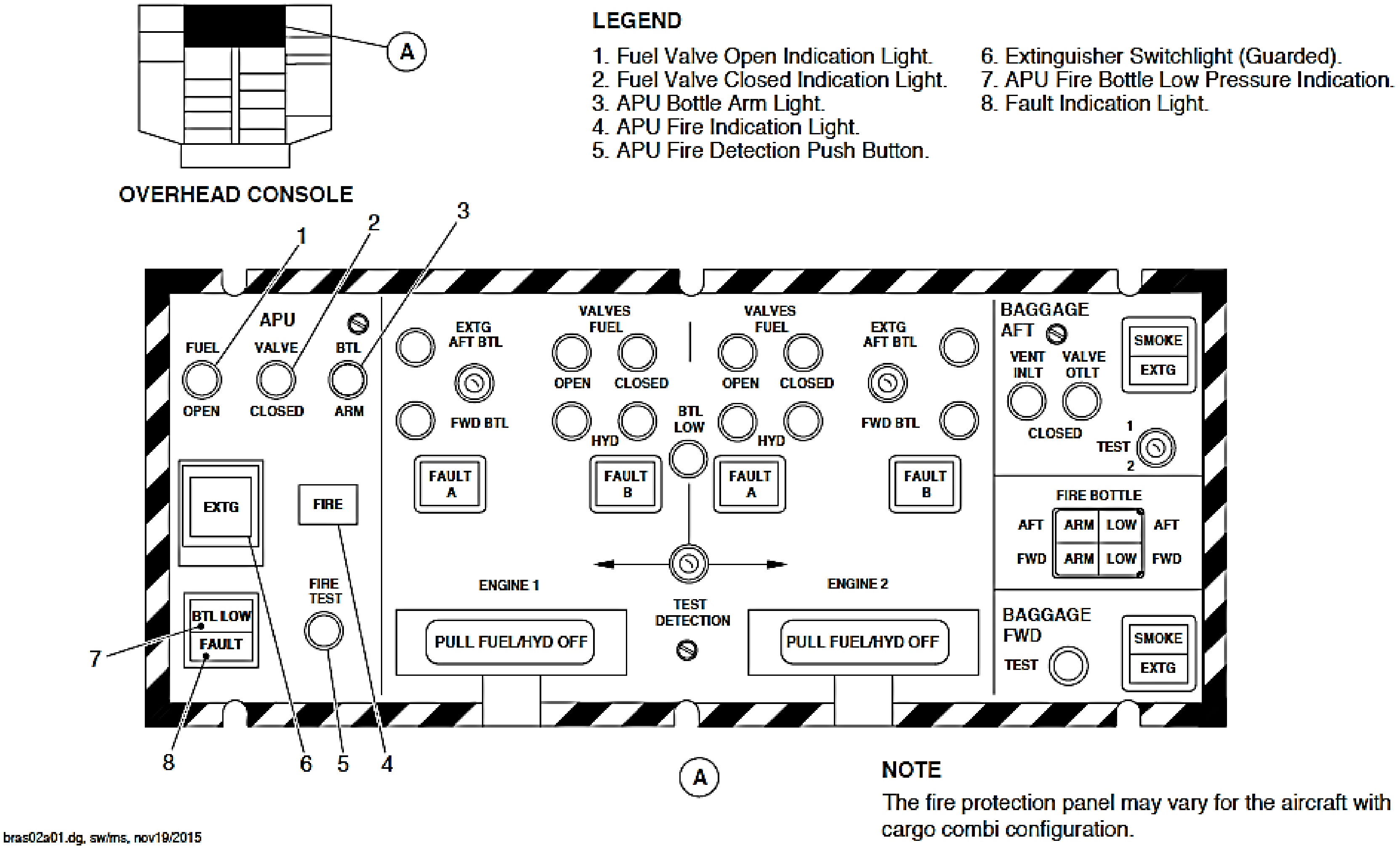


FIGURE - FIRE PROTECTION PANEL DETAIL FOR APU FIRE PROTECTION

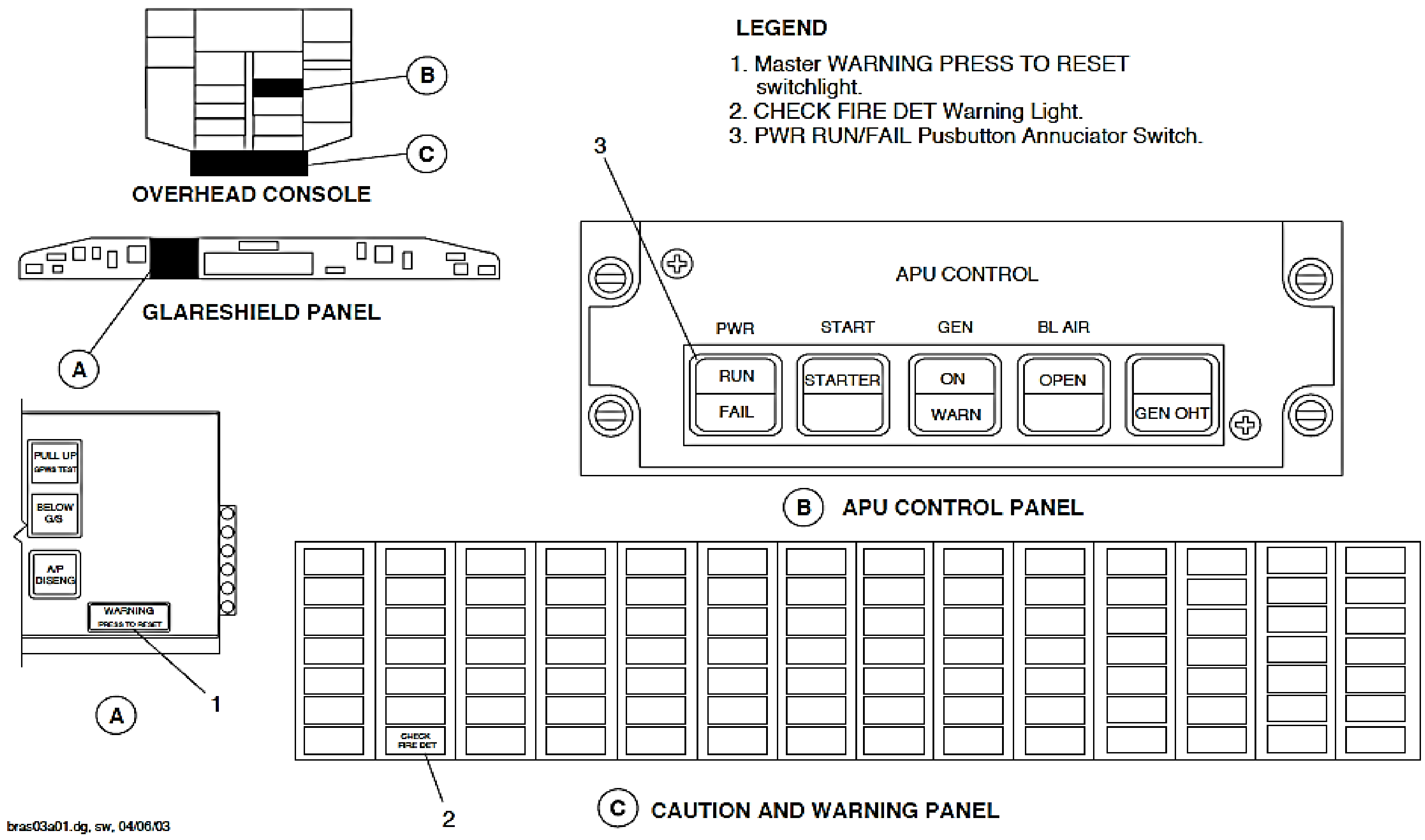


FIGURE - INDICATIONS FOR APU FIRE

APU FIRE EXTINGUISHING

Introduction

The Auxiliary Power Unit (APU) Fire Extinguishing System suppresses fire and overheat conditions in the APU.

General Description

Refer Figure - APU Fire Extinguisher Bottle

One fire extinguishing bottle is used to suppress fire and overheat conditions in the APU compartment. The APU fire extinguishing bottle has one fire extinguisher cartridge.

The APU fire extinguishing system has the components that follow:

- APU Fire Extinguisher Bottle
- APU Fire Extinguisher Cartridge.

Detailed Description

Refer Figure - Fire Protection Panel Detail for APU Fire Protection
Refer Figure - Indications for APU Fire
Refer Figure - APU Fire Detection and Extinguishing - Schematic

Indications on the flight compartment are used to monitor APU fire or overheat conditions.

No pilot action is required to operate APU fire protection. The control amplifier controls a relay which closes the APU fuel valve. When electrical power from the relay is supplied to the Full−Authority Digital−Electronic−Control (FADEC), the APU FUEL VALVE “CLOSED” advisory light on the Fire Protection Panel (FPP) comes on. The APU is shutdown automatically and the FAIL advisory light on the APU control panel comes on.

After a seven seconds delay, an electrical signal from the control amplifier ignites an Electro−Explosive Device (EED) in the extinguisher cartridge. The EED then explodes and the fire extinguishing agent is automatically released through the distribution tubes to the APU compartment. The APU BTL ARM light on the FPP goes out and the BTL LOW light comes on.

If automatic fire extinguishing fails and a fire or overheat condition is still sensed, a manual override capability is available by operating the APU EXTG switch located on the FPP. APU fire extinguisher bottle pressure is shown by a BTL LOW light on the FPP. The APU will automatically shut down.

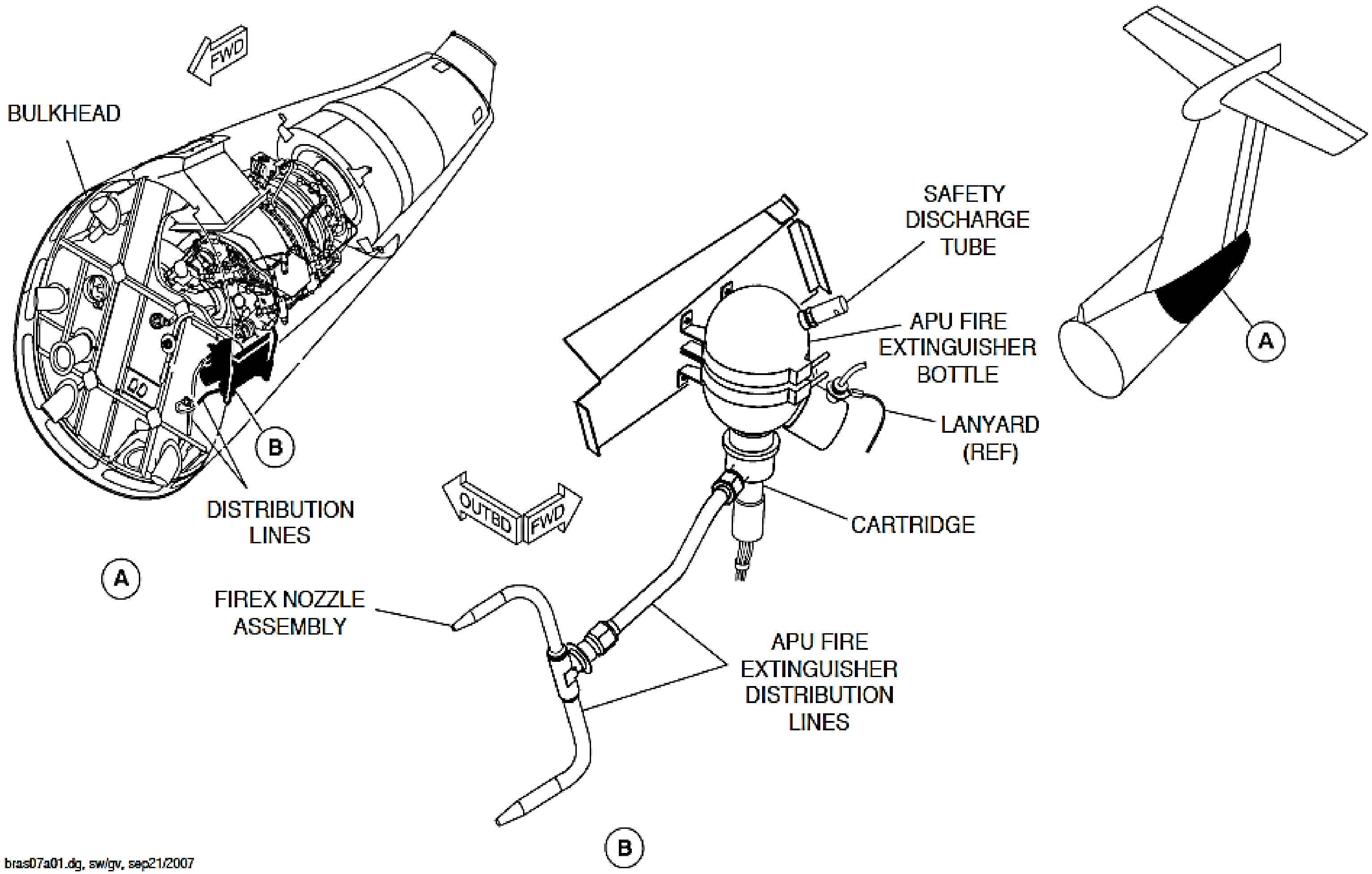


FIGURE - APU FIRE EXTINGUISHER BOTTLE

APU Fire Extinguisher Bottle

There is one APU fire extinguishing bottle for the APU compartment. The fire extinguishing bottle contains the suppressant necessary to combat a fire or overheat in the APU compartment.

The bottle is located in the aft equipment bay, forward of the APU compartment and is connected to the distribution tubes leading into the APU compartment.

The fire extinguisher is a stainless steel container filled with 4.37 lb. (1.98 kg) of liquid fire extinguishing agent Halon 1301 and pressurized with Nitrogen gas at 600 psig (4136 kPag).

On A/C with modsum 4Q126352 incorporated, the electrical connectors are attached with lanyards. These lanyards make sure that the correct connector is connected to the discharge squib during maintenance practices.

APU Fire Extinguisher Cartridge

APU Fire Extinguisher Cartridge causes the extinguisher bottle to discharge its contents through the distribution tubing leading to the APU compartment. There is one APU fire extinguisher cartridge attached to the discharge head, located on the bottom of the fire extinguishing bottle.

The cartridge contains a dual−bridgewire Electro−Explosive Device (EED) and redundant power lines with separate circuit breakers for reliability. Bridgewire “A” is taken from one circuit breaker and routed along the left side of the aircraft to the cartridge. Bridgewire “B” is taken from a separate circuit breaker and routed along the right side of the aircraft to the cartridge.

Malfunction of either power circuit breaker will not cause loss of fire extinguisher capability.

Each cartridge has an electrical connector. When an electrical signal is sent through the bridgewires, the EED ignites and explodes, rupturing a burst disk. This causes the extinguisher bottle to discharge its contents through the distribution tubes leading to the APU compartment.

The fire extinguisher cartridge must be handled with extreme care.

Discharge Nozzles

The APU fire extinguisher bottle is located outside of the APU compartment and is connected to distribution lines leading into the compartment.

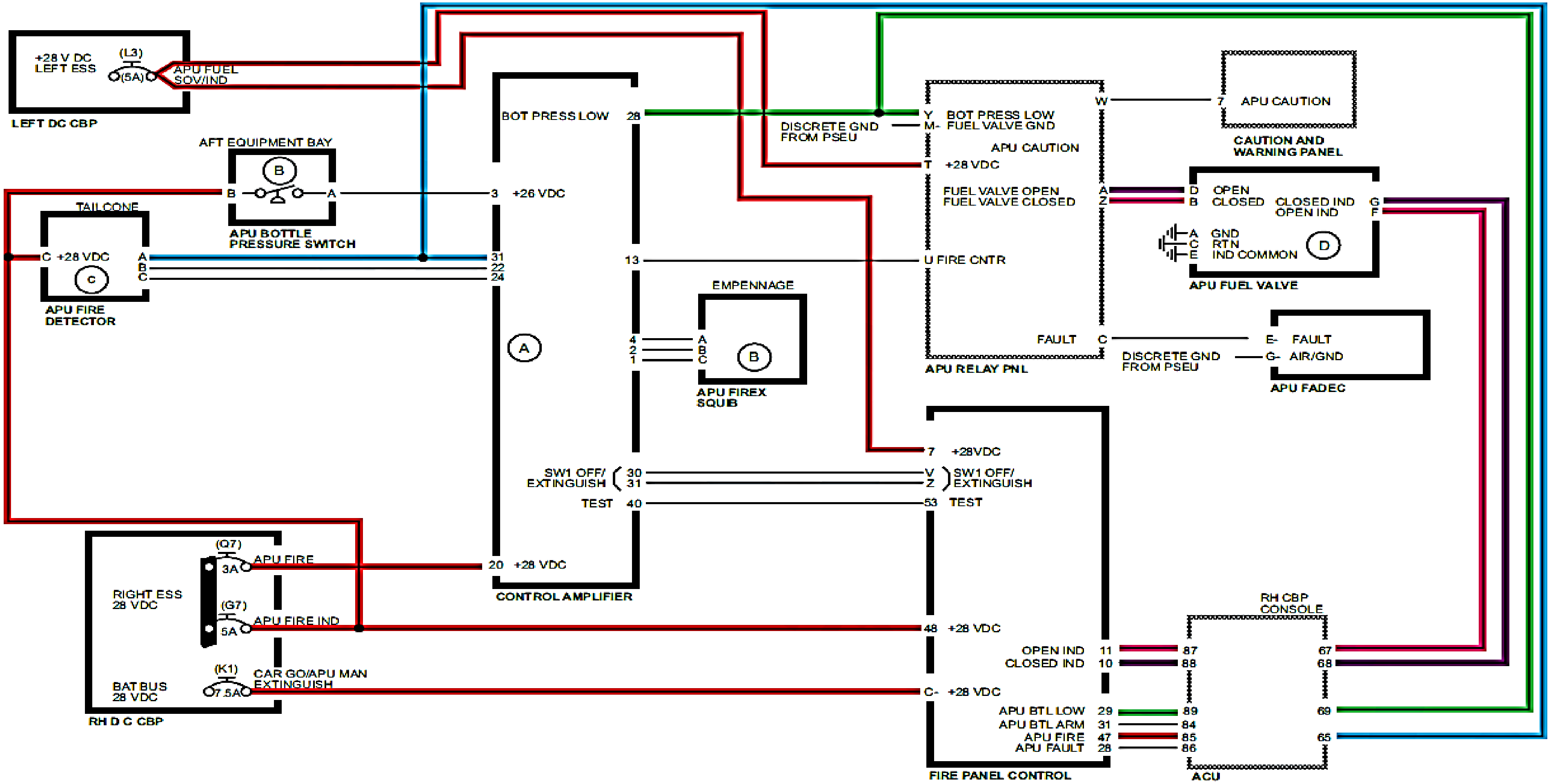


FIGURE - APU FIRE DETECTION AND EXTINGUISHING - SCHEMATIC

LAVATORY FIRE EXTINGUISHING

Introduction

The lavatory fire extinguisher is used to extinguish a possible fire in the lavatory waste−bin.

General Description

The lavatory compartment waste−bin is protected by a single, thermally−activated fire extinguisher with no electrical interface. If a fire occurs in the lavatory compartment waste−bin, the extinguishing agent is discharged into the bin.

Detailed Description

Refer Figure - Lavatory Compartment Waste−Bin Fire Extinguisher

The lavatory fire extinguisher has dual−discharge outlets that are thermally actuated with no electrical interface. When a fire occurs in the lavatory compartment waste−bin, the temperature of the end caps on the container increases. When the temperature reaches a set point, it causes the fusible seals to melt and release the end caps from the discharge tubes. The extinguishing agent is then released and discharged into the bin.

The lavatory fire extinguisher cannot be refilled or reused. It requires periodic weighing of the fire extinguisher bottle to make sure that it is full. If the discharge tubes are bent beyond the specified angle, fire suppression may not work properly.

Lavatory Fire Extinguisher

The Lavatory Fire Extinguisher is used to combat a fire in the lavatory waste−bin.

The lavatory fire extinguisher uses bromotrifluoromethane (Halon 1301) as the extinguishing agent. The bottle for the agent is a stainless steel sphere with an internal volume of 9 cubic inches (0.15 L). The bottle is charged with 0.25 pounds (0.113 kilograms) of agent to a pressure of 27 psi (186 kPa).

The lavatory fire extinguisher has two discharge outlets. If the end caps on the discharge tubes reach 174 °F (79 °C), the fusible alloy seals melt and release the end cap from the tubes. The extinguishing agent is then released into the waste−bin.

The lavatory fire extinguisher is located in the upper amenity assembly, just above the waste−bin. It is attached to the bulkhead with two screws, with the end caps protruding into the waste−bin.

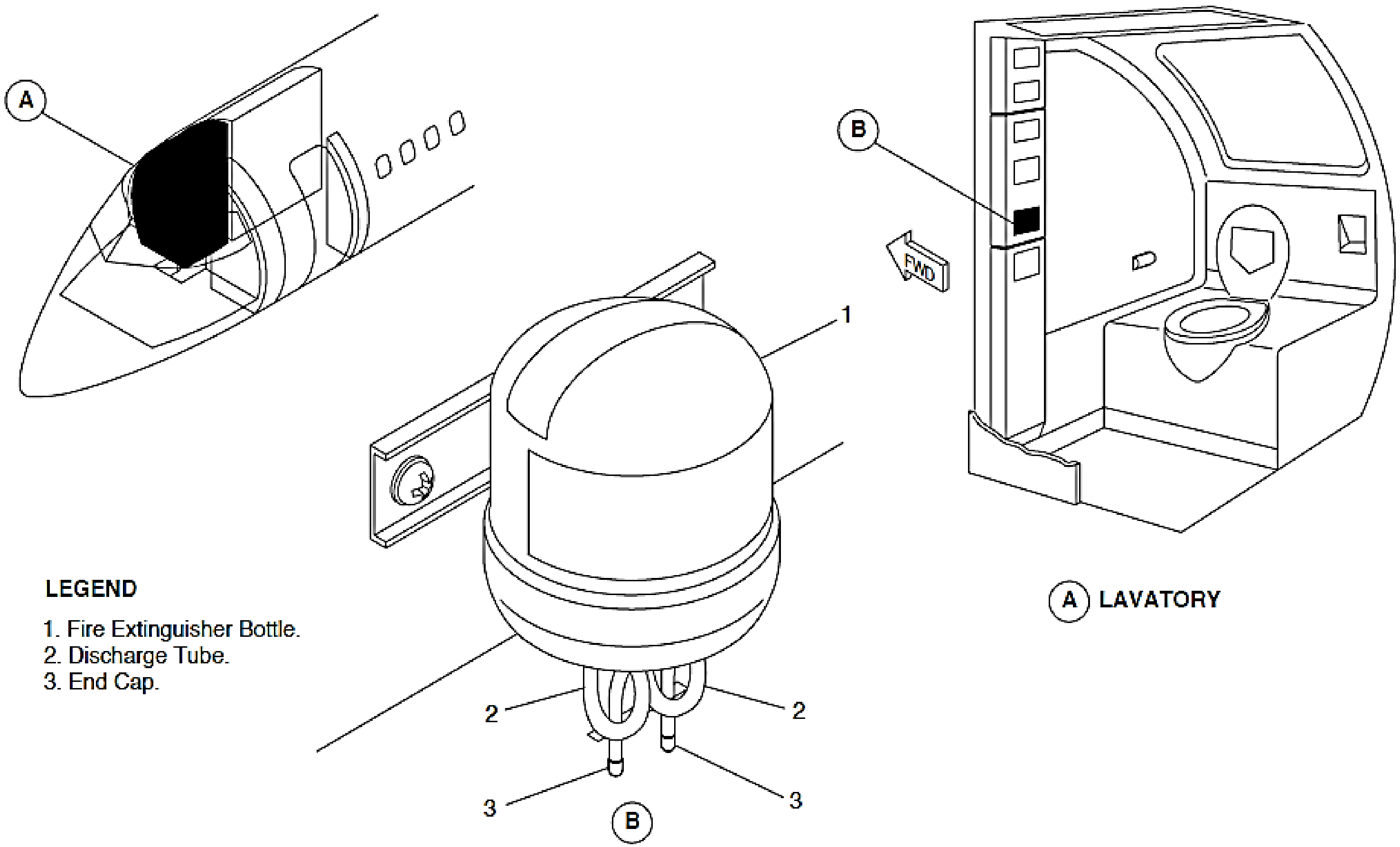


FIGURE - LAVATORY COMPARTMENT WASTE−BIN FIRE EXTINGUISHER

PORTABLE HAND–OPERATED FIRE EXTINGUISHERS

Introduction

The flight compartment and passenger compartment are protected by four portable hand–operated fire extinguishers.

General Description

Refer Figure - Aft Cabin Fire Extinguishers
Refer Figure - Hand Operated Flight Compartment Fire Extinguisher
Refer Figure - Hand Operated Fire Extinguisher - Aft Passenger Compartment
Refer Figure - Hand Operated Fire Extinguisher - Fwd Passenger Compartment

The portable hand–operated fire extinguishers are located at three locations in the aircraft. They are attached to the aircraft with quick–release brackets.

Four fire extinguishers are provided at the locations that follow:

- Flight compartment
- Forward passenger compartment
- Aft passenger compartment (2).

The location of the fire extinguishers may vary due to the different interior arrangements.

FIGURE - HAND OPERATED FLIGHT COMPARTMENT FIRE EXTINGUISHER

FIGURE - AFT CABIN FIRE EXTINGUISHERS

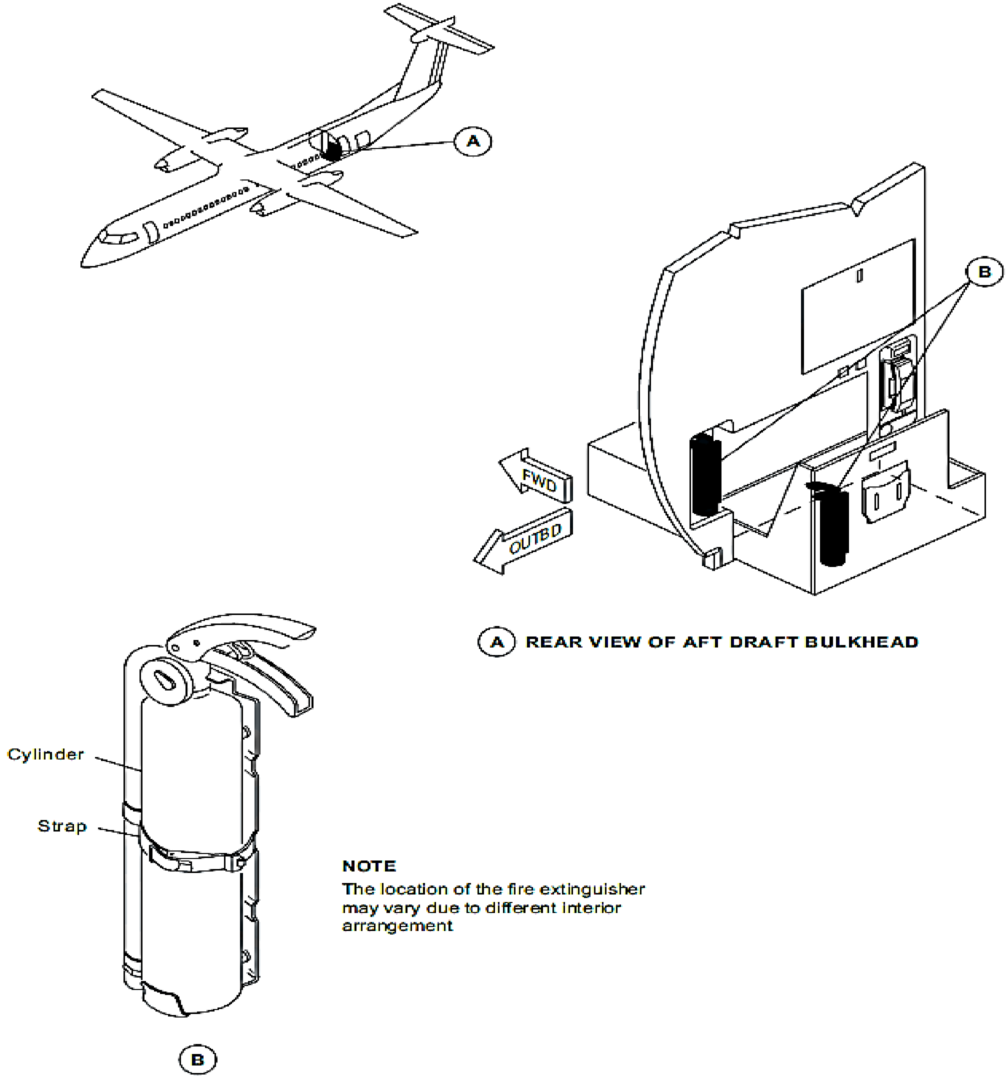


FIGURE - HAND OPERATED FIRE EXTINGUISHER - AFT PASSENGER COMPARTMENT

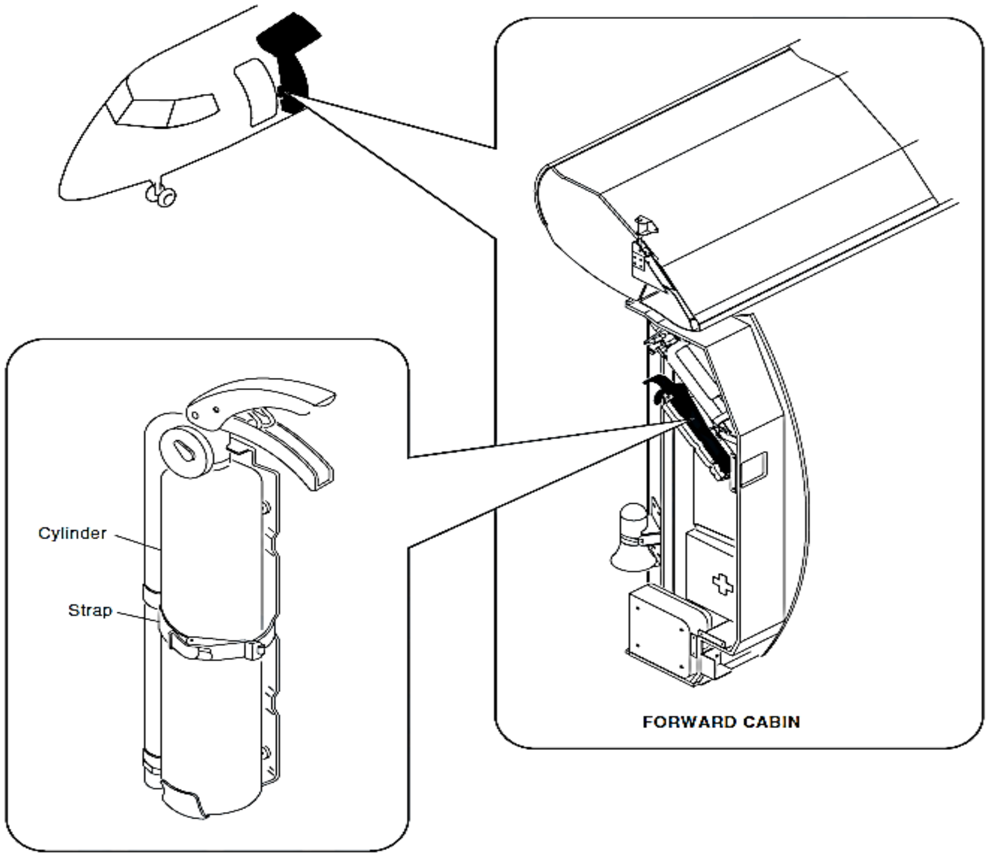


FIGURE - HAND OPERATED FIRE EXTINGUISHER - FWD PASSENGER COMPARTMENT